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Barth et al.

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(54) **HANDHELD POWER TOOL, IN PARTICULAR A SANDER, HANDHELD POWER TOOL DEVICE, PROTECTIVE DEVICE, HANDHELD POWER TOOL SYSTEM, AND METHOD FOR MANUFACTURING A HANDHELD POWER TOOL DEVICE**

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B24B 23/02 (2006.01)

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CPC **B25F 5/02** (2013.01); **B24B 23/005** (2013.01); **B24B 23/02** (2013.01)

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CPC B25F 5/02; B24B 23/005; B24B 23/02
See application file for complete search history.

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(57) **ABSTRACT**

A handheld power tool device for a handheld power tool, in particular a sander, preferably an orbital sander or a random orbital sander includes a housing unit, which has a handle housing and a drive housing for holding a drive unit. The drive housing defines a drive axis of the drive unit. The power tool device further includes a damping unit which couples the drive housing to the handle housing. The drive housing includes an extension arm, which extends at least radially with respect to the drive axis and on which at least a portion of the damping unit is arranged.

15 Claims, 10 Drawing Sheets

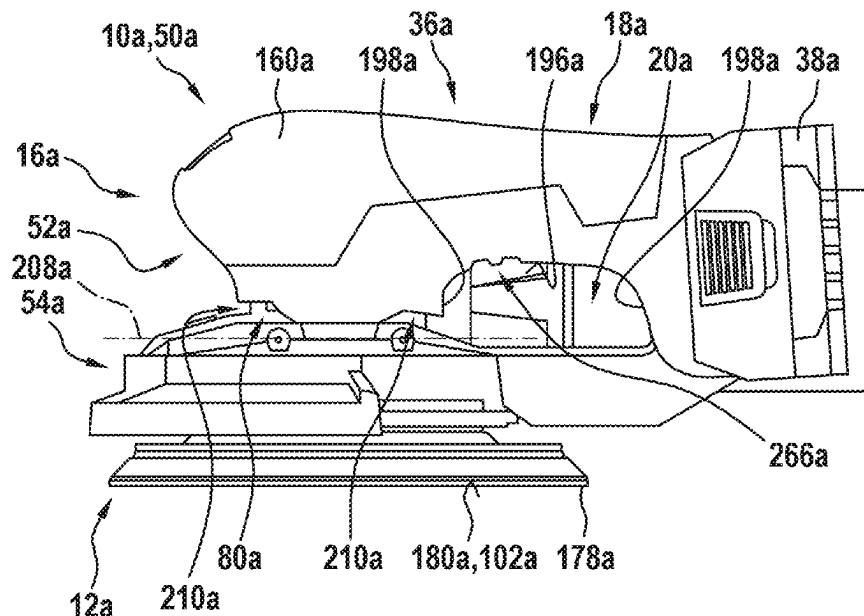


Fig. 1

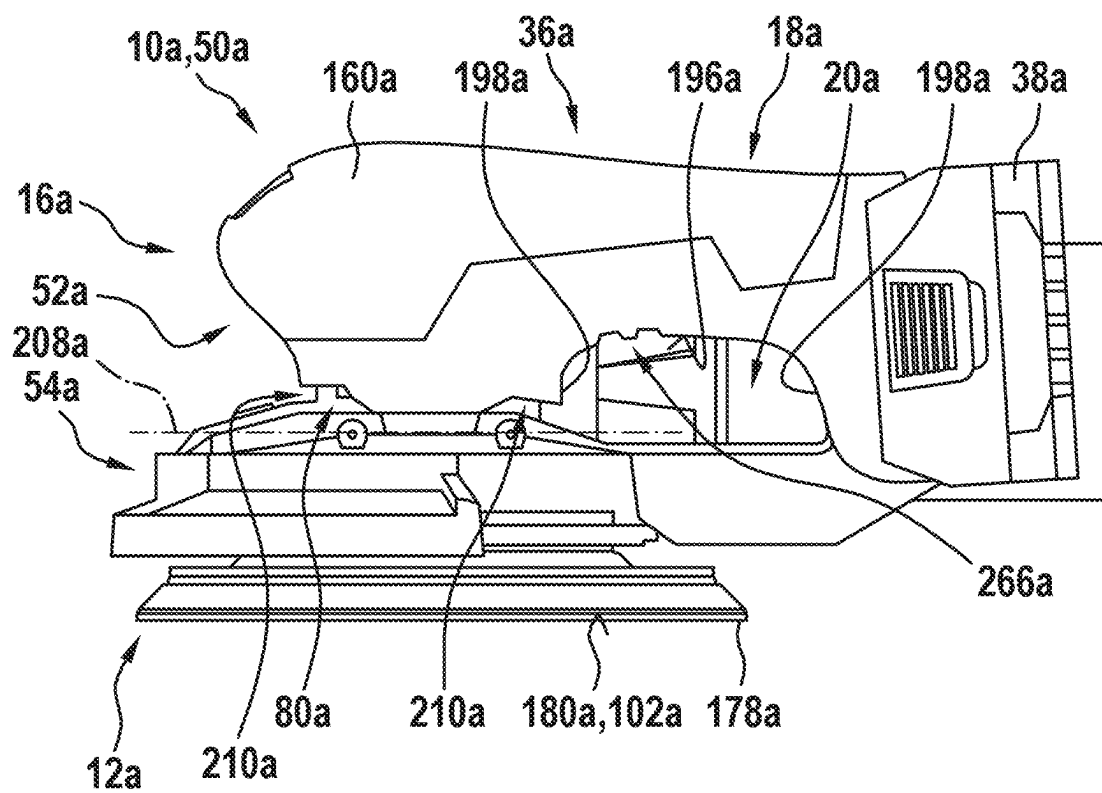


Fig. 2

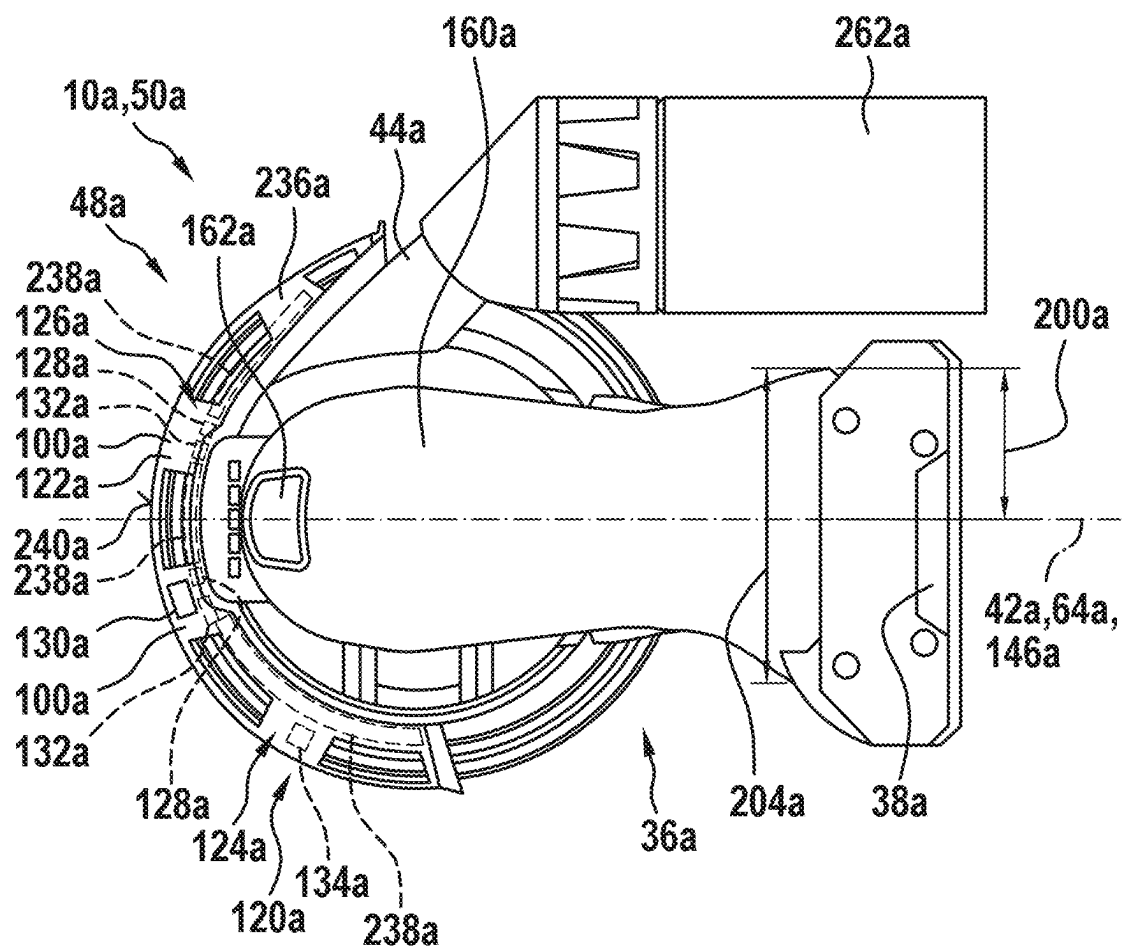
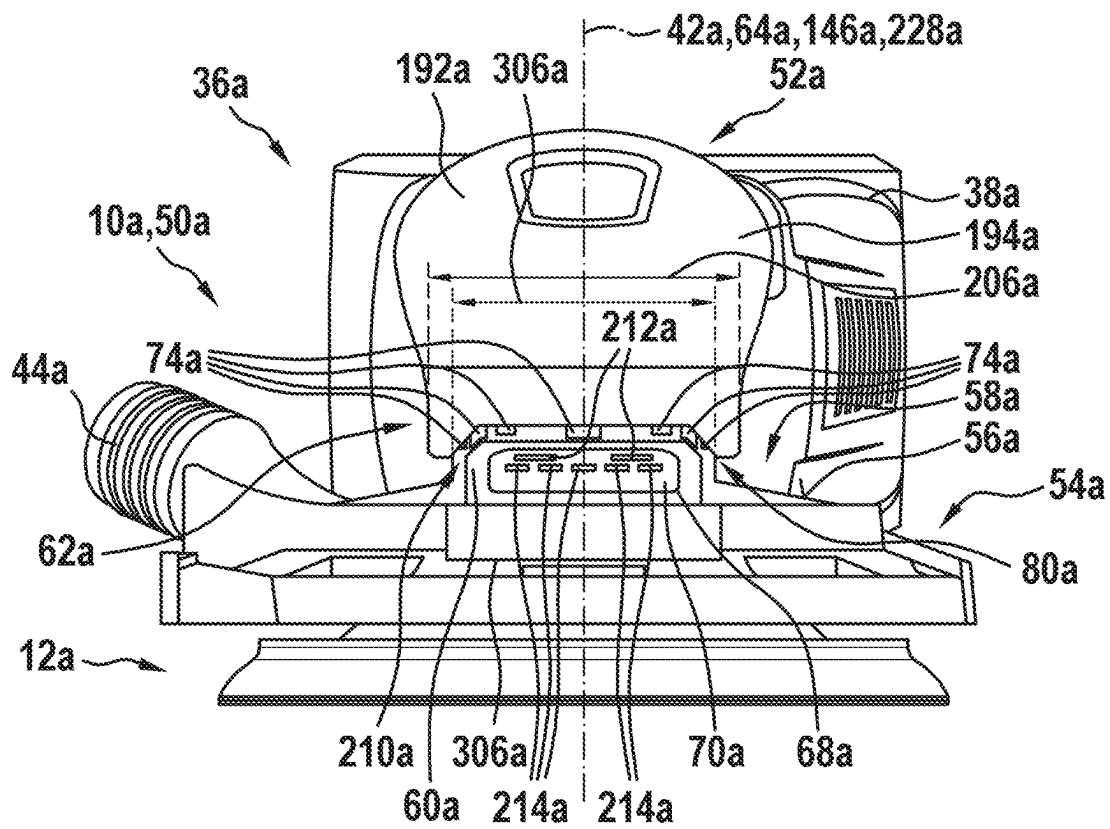
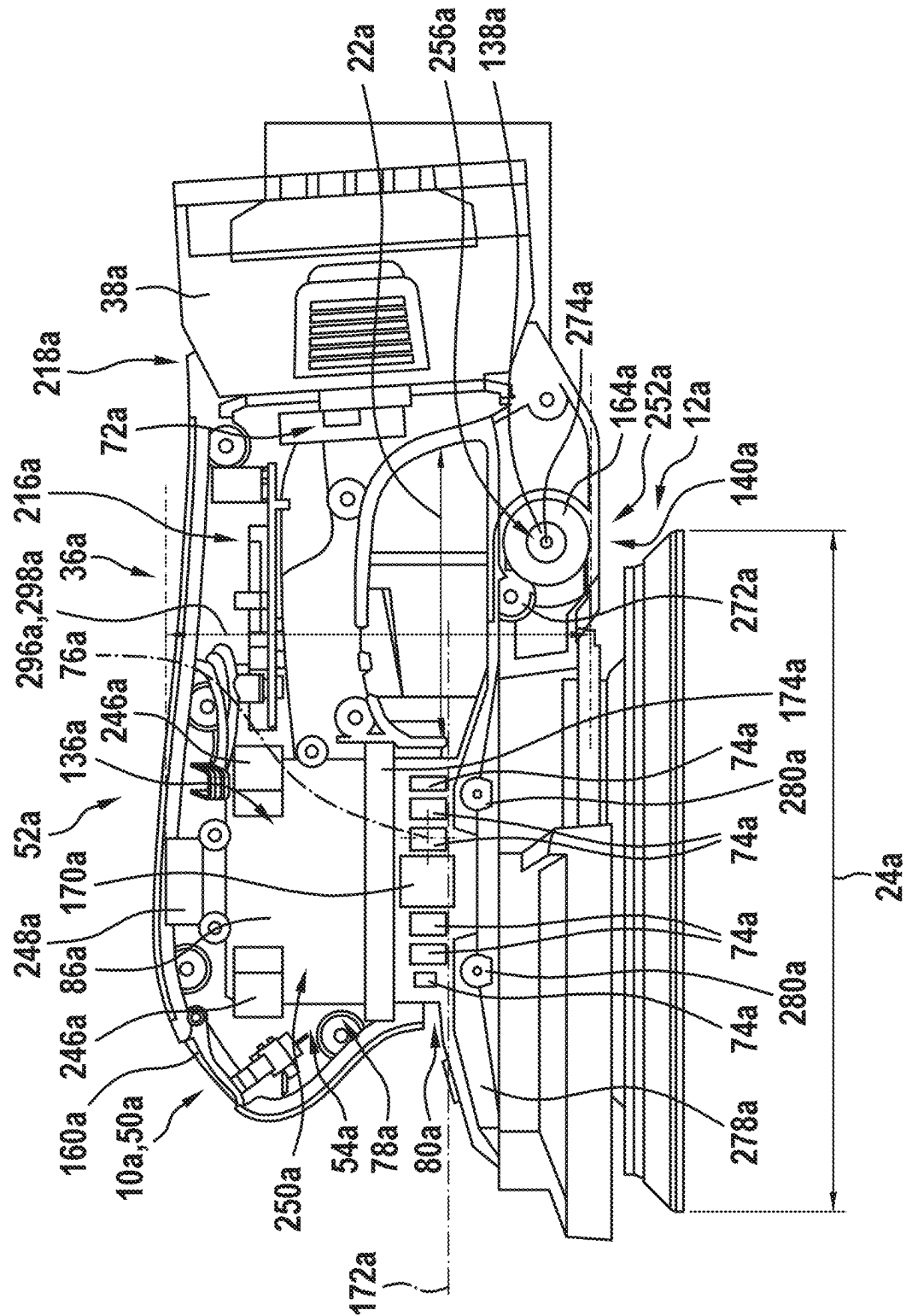


Fig. 3



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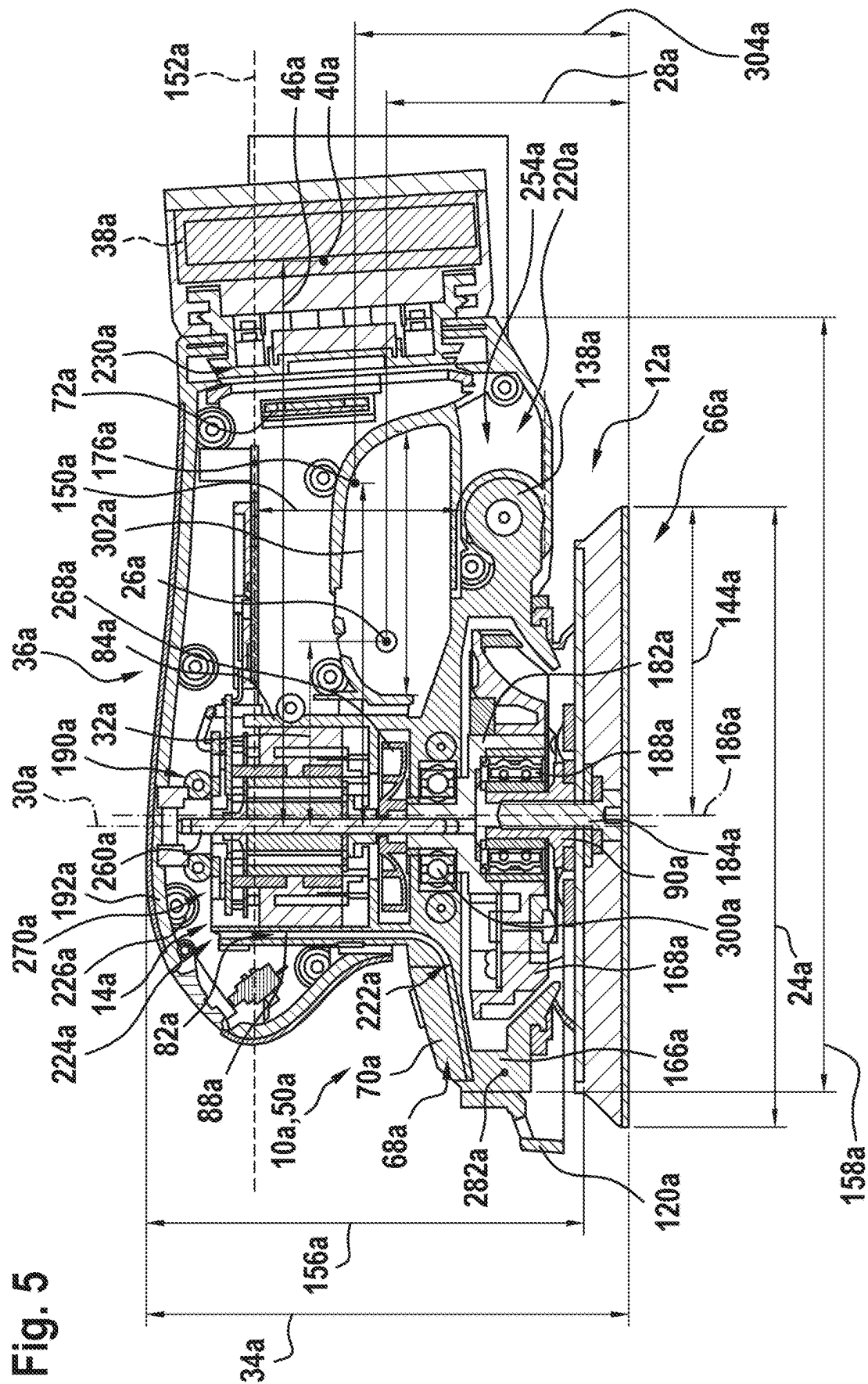


Fig. 6

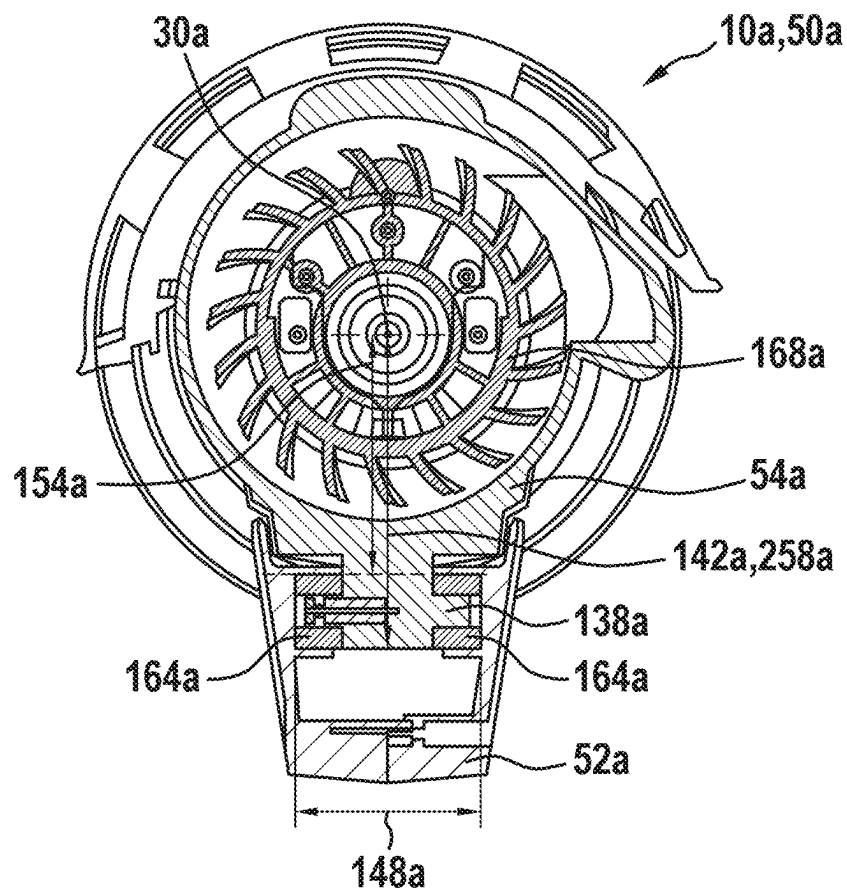


Fig. 7

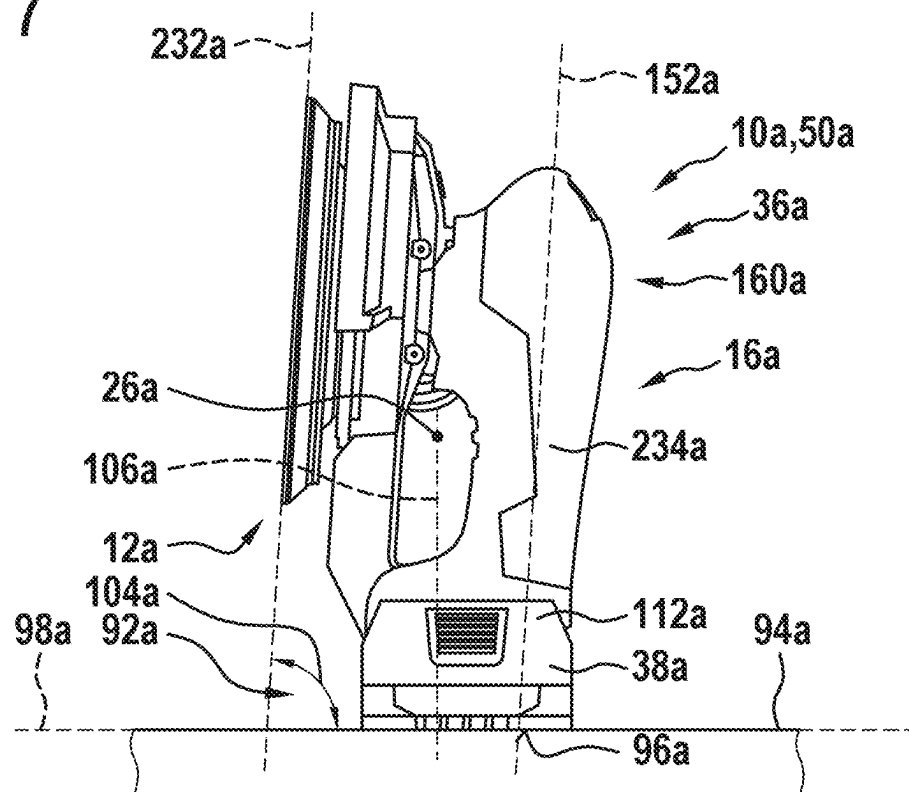


Fig. 8

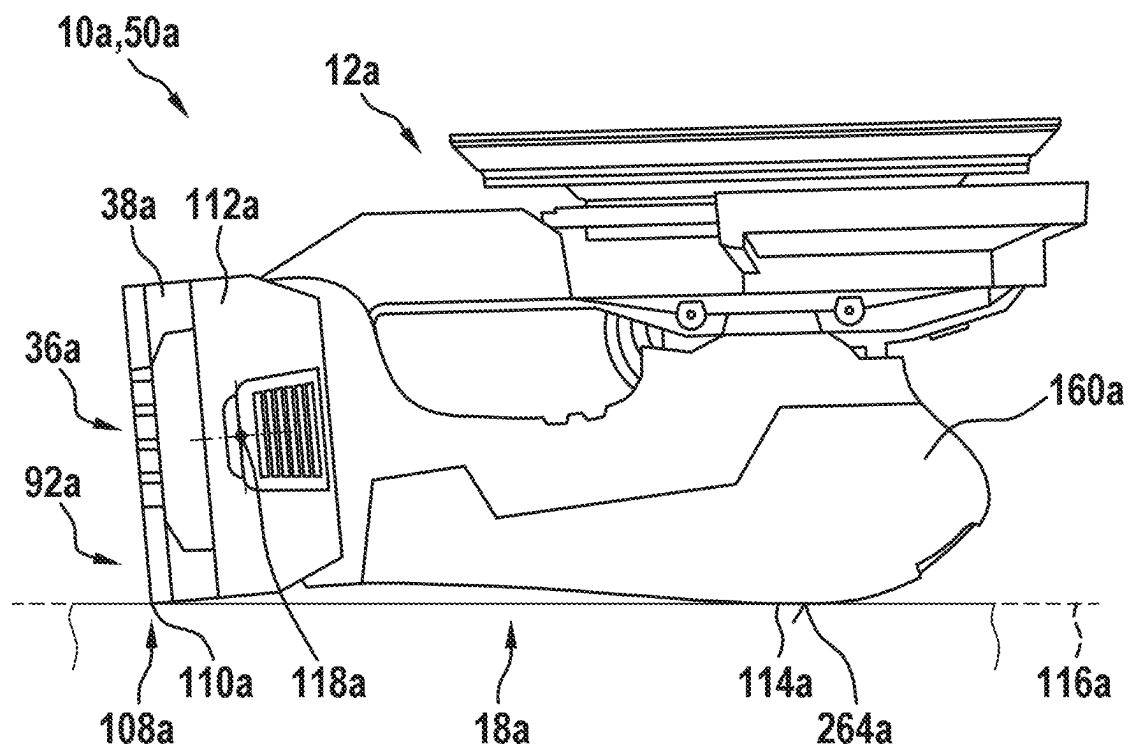


Fig. 9A

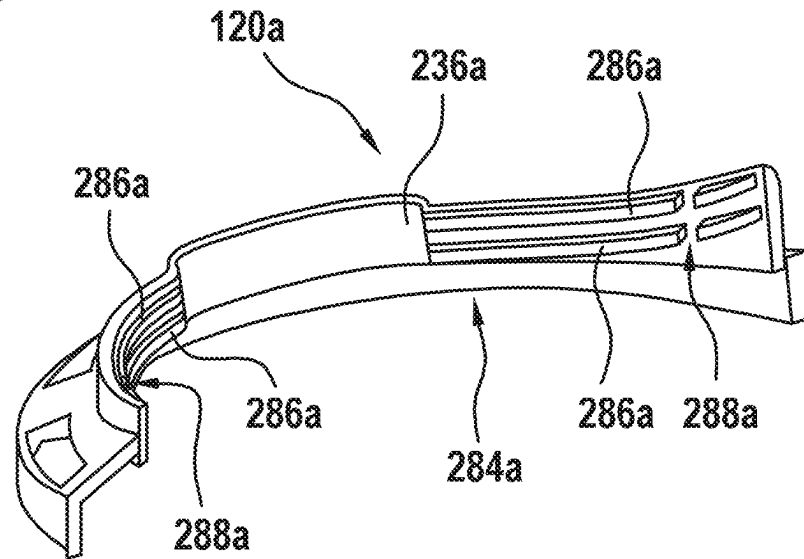


Fig. 9B

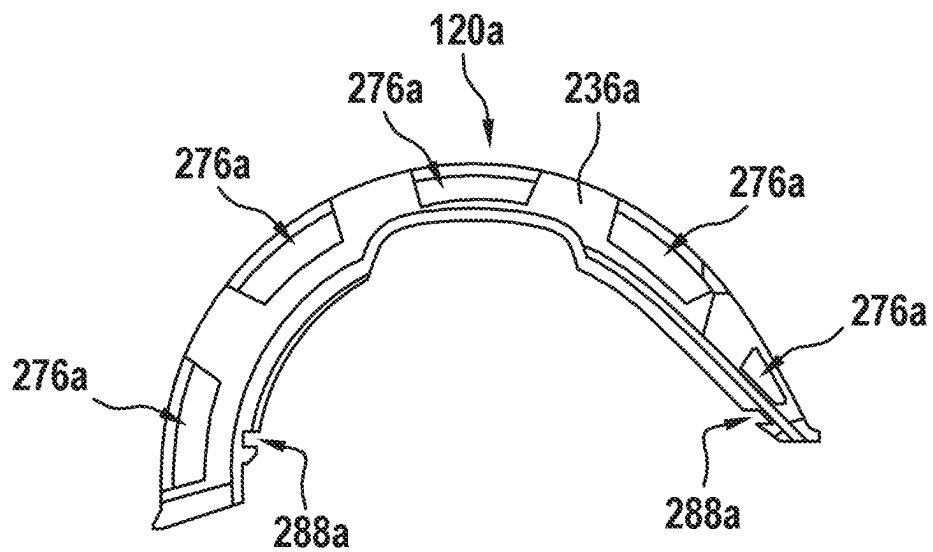


Fig. 10A

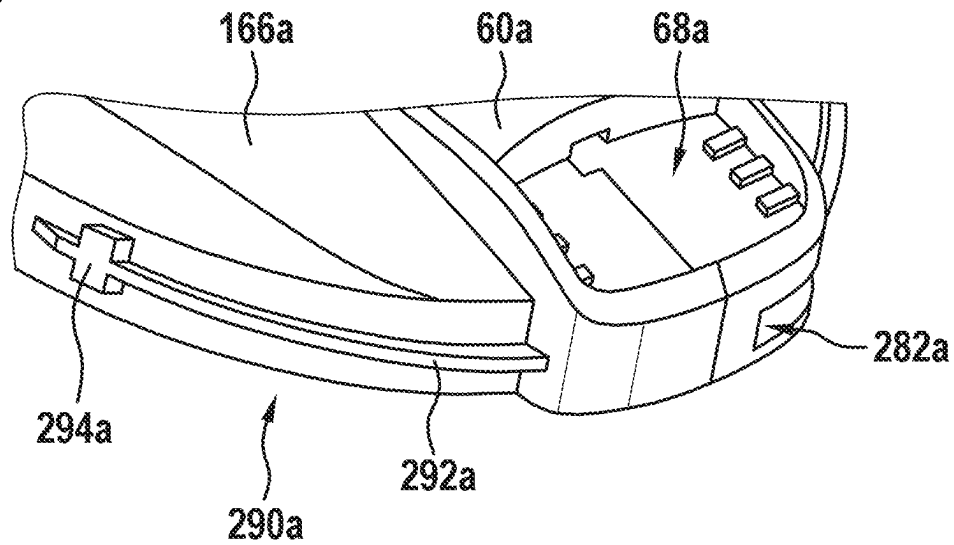


Fig. 10B

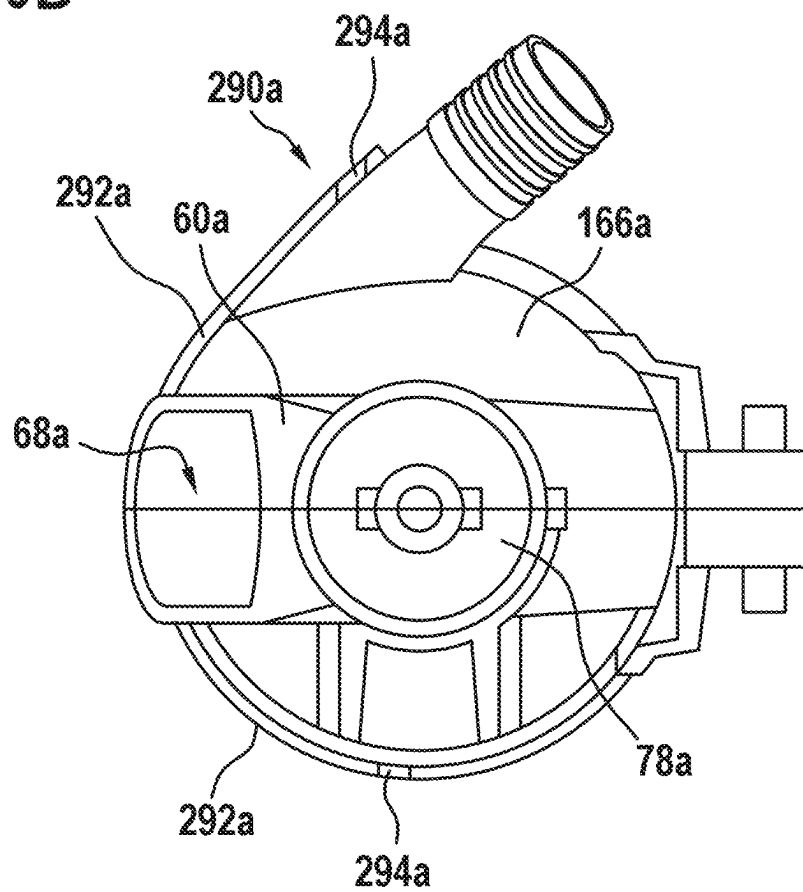


Fig. 11

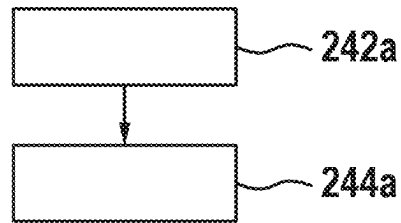
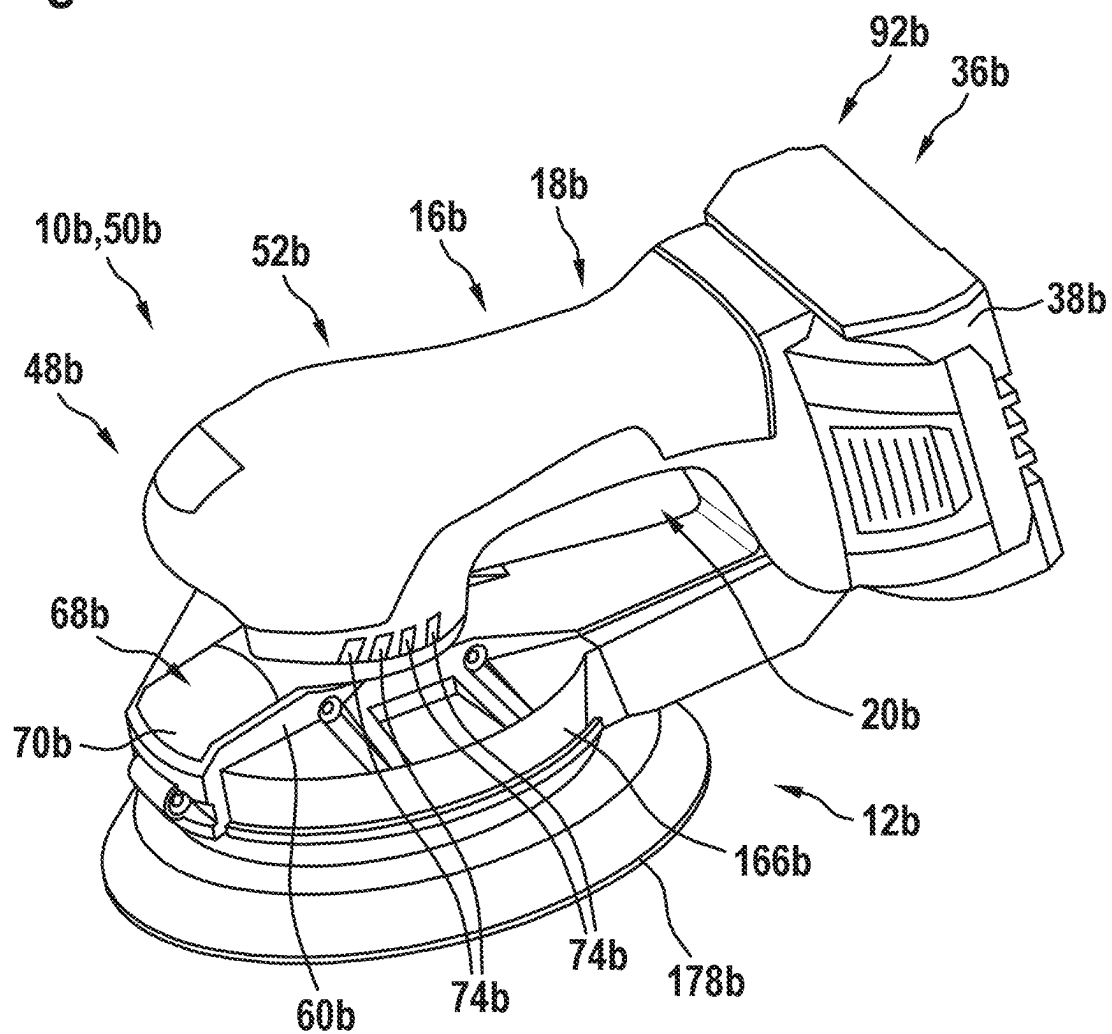


Fig. 12



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**HANDHELD POWER TOOL, IN
PARTICULAR A SANDER, HANDHELD
POWER TOOL DEVICE, PROTECTIVE
DEVICE, HANDHELD POWER TOOL
SYSTEM, AND METHOD FOR
MANUFACTURING A HANDHELD POWER
TOOL DEVICE**

This application claims priority under 35 U.S.C. § 119 to patent application no. DE 10 2023 205 363.2, filed on Jun. 9, 2023 in Germany, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

A handheld power tool device for a handheld power tool has already been proposed, comprising a housing unit which has a handle housing and a drive housing for holding a drive unit, with the drive housing defining a drive axis of the drive unit, said power tool also comprising a damping unit which couples the drive housing to the handle housing.

SUMMARY

The invention is based on a handheld power tool device for a handheld power tool, in particular a sander, preferably an orbital sander or a random orbital sander comprising a housing unit which has a handle housing and a drive housing for holding a drive unit, whereby the drive housing defines a drive axis of the drive unit, said power tool also comprising a damping unit that couples the drive housing to the handle housing.

It is proposed that the drive housing comprises an extension arm, which extends at least radially with respect to the drive axis and on which at least a portion of the damping unit is arranged. Advantageously, a particularly efficient decoupling of the handle housing from vibrations on the drive housing can be achieved using simultaneously particularly precise and/or comfortable handheld power tool guidance. A particularly high level of operating convenience can be achieved. A particularly wide support of the drive housing on the handle housing can be advantageously achieved. A particularly stable guidance for the handheld power tool can be achieved in an advantageous manner. A particularly high level of operating safety can be achieved.

In particular, a motor housing of the drive unit defines the drive axis. Preferably, the motor housing comprises a motor holder, in the central longitudinal axis of which a motor axis comes to rest. The drive axis is defined in particular by the motor axis. In particular, the central longitudinal axis is congruent with the drive axis.

The damping unit preferably comprises multiple damping elements. The damping elements are preferably made of at least one elastic material, in particular an elastic plastic. The damping elements can be designed as blocks, rings, ring segments, plates, or the like. The drive housing is preferably connected to the handle housing via the damping elements. The damping unit is in particular intended to decouple the handle housing from the drive housing from vibrations. The handle housing is preferably only connected to the drive housing via the damping unit, in particular the damping elements of the damping unit.

At least a portion of the damping elements can preferably be fixed between the drive housing and the handle housing by means of a clamp. It is also conceivable that the damping elements comprise an adhesive layer for attachment to the drive housing and/or the handle housing. Additionally or

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alternatively, it is conceivable that at least a portion of the damping elements can be attached to the handle housing and/or the drive housing by means of damping element holders, e.g. recesses, holding bolts, or the like, and/or by means of a screw connection or the like.

In particular, the handle housing is coupled to the drive housing via two different coupling regions of the damping unit. Preferably, the handle housing comprises two inner regions, in particular at a distance from each other, within each of which a portion of the drive housing is arranged. In particular, the two coupling regions are arranged in the two inner regions.

The extension arm is preferably arranged on a side of the drive housing facing a battery pack interface of the sander. The extension arm is preferably arranged inside the handle housing, in particular in an inner region of the two inner regions of the handle housing. The extension arm forms an outermost point of the drive housing, particularly when viewed in a direction perpendicular to the drive axis and along a separating plane of the handle housing. The extension arm is preferably at least partially enclosed by the handle housing.

The portion of the damping unit arranged on the extension arm in particular comprises at least one damping element of the damping elements. The extension arm preferably comprises at least one damping element holder for the at least one damping element. The shape of the damping element is preferably adapted to the damping element holder. The damping element holder is, e.g., designed as a holding bolt, in particular for holding a ring-shaped damping element. The damping element holder is, e.g., intended to hold the at least one damping element perpendicular to the main extension plane of the handle housing, in particular to a main extension plane of a bow-shaped handle of the housing unit, and/or to the separating plane. The “main extension plane” of a structural unit or an element can be understood as a plane which is parallel to a largest side surface of a smallest possible imaginary cuboid, which just completely encloses the structural unit, and in particular extends through the center of the cuboid. The extension arm is, e.g., designed to be T-shaped in particular to hold at least two damping elements. Alternatively, however, it is also conceivable that the damping element holder is, e.g., designed as a recess, in particular for holding a damping element designed as a block, plate or the like. It is conceivable that the at least one damping element is fixed to the damping element holder by the handle housing, in particular handle housing shells of the handle housing. Alternatively or additionally, however, it is also conceivable that the damping element is glued, screwed, or the like to the damping element holder.

In one aspect of the invention, which can be considered in particular on its own as well as in conjunction with other aspects of the invention, the invention is based on a sander, in particular a battery-operated sander, having a tool holder, in particular a sanding pad, having a drive unit, in particular the drive unit specified hereinabove for driving the tool holder and having a housing unit, in particular the housing unit specified hereinabove, which comprises a bow-shaped handle, in particular the bow-shaped handle specified hereinabove. It is proposed that a ratio of a maximum longitudinal extent of a handle recess of the bow-shaped handle to a maximum longitudinal extent of the tool holder is at least 0.35 and at most 0.5. A sander with particularly favorable ergonomic properties can be provided in an advantageous manner. Advantageously, particularly depending on the size of the tool holder, a particularly large amount of space can be provided for an operator's fingers in the handle recess for

guiding the sander. Particularly precise and/or safe handling of the sander can be achieved. A particularly high level of operating convenience can be achieved in an advantageous manner.

The sander is preferably designed as a battery-operated sander. Alternatively, however, it is also conceivable that the sander is designed as a grid-powered sander. The sander is preferably designed as a random orbital sander. Alternatively, however, it is also conceivable that the sander is designed as an orbital sander, a delta sander, a flower sander, or the like.

The tool holder is preferably designed as a sanding pad. The tool holder is preferably designed as a round sanding pad. Alternatively, however, it is also conceivable that the tool holder is designed as a delta pad, as a square, preferably rectangular, sanding pad or as another sanding pad that a skilled person considers useful. In particular, the tool holder comprises fastening means for fastening a tool, preferably a sanding means. The fastening means comprise, e.g., one or multiple Velcro surfaces, one or multiple clips, a combination of these, or other fastening means that a skilled person considers useful. The sanding means is preferably designed as an abrasive paper, an abrasive fleece, an abrasive grid, or the like.

In particular, the drive unit comprises a drive element, preferably an eccentric. The drive element is preferably intended to be driven around a drive axis of the drive unit. The sander preferably comprises an output unit. In particular, the tool holder is a portion of the output unit. The output unit preferably comprises a connecting piece. The connecting piece is arranged on the drive element, preferably attached. The tool holder is preferably attached to the connecting piece, in particular non-rotatably, e.g. by means of a screw or the like. In particular, the output element has an output axis. The output axis preferably extends at least substantially parallel to the drive axis. In particular, the drive axis is at a distance from the output axis. The term “substantially parallel” can in this context be understood to mean an orientation of a direction relative to a reference direction, in particular in a plane, whereby the direction has a deviation relative to the reference direction that is in particular less than 8°, advantageously less than 5°, and particularly advantageously less than 2°. In particular, the connecting piece and/or the tool holder, preferably the output axis, moves in at least one operating state around the drive axis, preferably in a circular motion. In particular, the drive element is intended to move the connecting piece and/or the tool holder, preferably the output axis, in a preferably circular motion about the drive axis of the tool holder.

The term “intended” should be understood to mean specially furnished, specially designed, and/or specially equipped. An object being “intended” for a specific function is understood to mean that the object fulfills and/or executes this specific function in at least one application and/or operating state.

The connecting piece is preferably attached to the drive element so that it can rotate, in particular about the output axis. The sander preferably comprises at least one bearing element. In particular, the connecting piece is mounted on the drive element so that it can rotate via the bearing element, preferably about the output axis. The bearing element is preferably arranged between the drive element and the connecting piece. The bearing element is preferably designed as a radial bearing, in particular as a rolling bearing, e.g. as a ball bearing, or as a plain bearing.

The drive unit is preferably intended to drive the output unit, in particular the tool holder.

The drive unit preferably comprises an electric motor of the like, in particular for driving the drive element.

The sander preferably comprises a housing unit. The housing unit preferably comprises a drive housing to hold the drive unit. The drive housing preferably comprises a fan housing to hold an extraction fan. Preferably, the extraction fan is provided to extract the material removed from the workpiece during machining and, in particular, to blow it out of a dust outlet of the sander. The drive housing preferably comprises the motor housing for holding the drive unit. The fan housing is preferably arranged between the motor housing and the tool holder. The extraction fan is preferably made of metal, in particular cast. Alternatively, however, it is also conceivable that the extraction fan is made of plastic or a combination of plastic and metal.

In particular, the housing unit comprises a handle housing. The bow-shaped handle is preferably a portion of the handle housing. The handle housing preferably comprises two handle housing shells, in particular those specified hereinabove, which are connected to each other, in particular in the separating plane of the handle housing. The handle housing is preferably arranged separately from the drive housing, in particular at a distance from the drive housing. In particular, the handle housing comprises a shell structure, with the two handle housing shells acting as half shells.

The bow-shaped handle preferably comprises a finger grip surface. The bow-shaped handle preferably comprises two walls, in particular side walls, which delimit the finger grip surface, in particular in the main extension plane of the bow-shaped handle and viewed in a direction perpendicular to the drive axis. The bow-shaped handle is preferably designed as a closed bow-shaped handle. Alternatively, however, it is also conceivable that the bow-shaped handle is designed as an open bow-shaped handle. Preferably, the walls define, in particular limit, the handle recess at least partially.

The maximum longitudinal extent of the handle recess preferably extends at least substantially parallel to a tool holder surface of the tool holder. The maximum longitudinal extension of the handle recess preferably extends in the main extension plane of the bow-shaped handle, preferably in a main extension plane of the handle housing, in particular in the separating plane of the handle housing. Preferably, the main extension plane of the bow-shaped handle corresponds to the main extension plane of the handle housing. Alternatively, it is conceivable that the main extension plane of the bow-shaped handle is different from the main extension plane of the handle housing. The separating plane is preferably congruent with the main extension plane of the handle housing and/or the main extension plane of the bow-shaped handle. Alternatively, however, it is also conceivable that the main extension plane of the bow-shaped handle and/or the main extension plane of the handle housing are/is different from the separating plane.

The tool holder surface preferably extends at least substantially perpendicular to the separating plane, the main extension plane of the handle housing, and/or the main extension plane of the bow-shaped handle. The term “substantially perpendicular” can be understood to mean an orientation of a direction relative to a reference direction, whereby, in particular viewed in a projection plane, the direction and the reference direction enclose an angle of 90° and the angle has a maximum deviation of in particular less than 8°, advantageously less than 5°, and particularly advantageously less than 2°.

In particular, the maximum longitudinal extent of the tool holder extends at least substantially parallel to the maximum

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longitudinal extent of the handle recess and/or the tool holder surface. The maximum longitudinal extension of the tool holder preferably extends along the separating plane of the handle housing, the main extension plane of the handle housing, and/or along the main extension plane of the bow-shaped handle. The maximum longitudinal extension of the tool holder is defined in particular by a maximum longitudinal extension of the tool holder surface. In particular, the tool is intended to be arranged on the tool holder surface.

Preferably, the ratio of the maximum longitudinal extent of the handle recess of the bow-shaped handle to the maximum longitudinal extent of the tool holder is at least 0.38. Preferably, the ratio of the maximum longitudinal extent of the handle recess of the bow-shaped handle to the maximum longitudinal extent of the tool holder is at most 0.42. The ratio of the maximum longitudinal extent of the handle recess of the bow-shaped handle to the maximum longitudinal extent of the tool holder is at least substantially 0.4. The maximum longitudinal extent of the handle recess is preferably at least 50 mm, preferably at least 55 mm and particularly preferably at least 60 mm. The maximum longitudinal extent of the handle recess is preferably at most 80 mm, preferably at most 70 mm, and particularly preferably at most 75 mm. Particularly preferably, the maximum longitudinal extent of the handle is at least substantially 61 mm. Alternatively, however, it is also conceivable that the maximum longitudinal extent of the handle recess is greater than 80 mm or less than 50 mm.

In particular, the handle housing comprises a palm grip, in particular a grip knob. The palm grip is preferably arranged such that the drive axis intersects the palm grip. The bow-shaped handle merges into the palm grip in particular. In particular, the fan housing is located between the tool holder and the palm grip. The handle housing, in particular the palm grip, surrounds the motor housing, preferably at least in a plane perpendicular to the drive axis.

The sander preferably comprises the battery pack interface for electrical and/or mechanical connection to a battery pack. In particular, the battery pack interface comprises guidance means, e.g. one or multiple guide rails or the like, for guiding the battery pack during assembly and/or disassembly at the battery pack interface. The battery pack interface, in particular the guidance means, preferably specifies an insertion direction for the battery pack when the battery pack is mounted on the battery pack interface. The insertion direction preferably extends at least substantially perpendicular to the drive axis, to the separating plane, to the main extension plane of the bow-shaped handle, and/or to the main extension plane of the handle housing. In particular, the insertion direction extends at least substantially parallel to the tool holder surface. Preferably, the battery pack interface is arranged on the handle housing, in particular on a side of the bow-shaped handle facing away from the palm grip. In particular, the battery pack interface has no interface with the drive axis and/or a main extension plane of the tool holder surface.

Preferably, the handheld power tool, in particular the sander, comprises at least one input element. Preferably, the handheld power tool can be switched on and/or off by the operator actuating the input element. Alternatively or additionally, it is conceivable that the input element can be used to adjust the rotational speed of the handheld power tool or the like. The input element can, e.g., be designed as a button, a rotary wheel, a switch, or the like. The input element is, e.g., arranged on the palm grip, preferably on a side of the

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palm grip facing away from the battery pack interface, the tool holder, and/or the bow-shaped handle.

The sander is preferably intended for one-handed operation, in particular by means of the bow-shaped handle and the palm grip. Alternatively, however, it is also conceivable that the sander is designed as a two-handed guide, in which case one hand of the operator is arranged on the palm grip and one on the bow-shaped handle.

In one aspect of the invention, which can be considered in particular on its own as well as in conjunction with other aspects of the invention, the invention is based on a hand-held power tool device, in particular a hand-held sander device, for a handheld power tool, in particular a sander, preferably the sander specified hereinabove, comprising a housing unit, in particular the housing unit specified hereinabove, which comprises at least one handle housing, in particular the handle housing specified hereinabove, and a drive housing, in particular the drive housing specified hereinabove, which is coupled to the handle housing. It is proposed that the drive housing comprises, on a surface of an upper side of the drive housing, a center bar which is raised relative to the surface and above which, in particular, one end of the handle housing is arranged, whereby the center bar slopes down onto the surface in both directions, starting from a main extension plane of the handle housing, in particular the main extension plane of the handle housing specified hereinabove, and perpendicular to the main extension plane of the handle housing.

Advantageously, a design of the drive housing with such a center bar enables a particularly space-saving design of the handheld power tool device, in particular with regard to the overall height of the handheld power tool. A particularly large gripping surface can be realized on the handle housing. Contact between an operator's fingers and the drive housing can be efficiently counteracted in an advantageous manner. The operator can be protected particularly efficiently against vibrations. A particularly high level of operating convenience can be achieved in an advantageous manner. Particularly precise and/or safe guidance of the handheld power tool can be achieved.

The center bar stands out in particular from the adjacent surface. A maximum transverse extent of the center bar is preferably smaller than a maximum overall transverse extent of the handle housing. The maximum transverse extent of the center bar preferably extends at least substantially perpendicular to the main extension plane of the bow-shaped handle, the main extension plane of the handle housing, the separating plane, and/or the drive axis. In particular, the maximum transverse extent of the center bar extends at least substantially parallel to the tool holder surface of the tool holder. The maximum transverse extent of the center bar is preferably smaller than a maximum transverse extent of the handle housing at the end of the center bar arranged above the center bar. The end of the handle housing is arranged above the center bar, particularly in relation to the tool holder and/or the surface.

The center bar preferably comprises a main extension axis. A "main extension axis" of an object can be understood in particular as an axis that extends parallel to a longest edge of a smallest geometric cuboid that just completely encloses the object, and in particular extends through the center of the cuboid. In particular, the main extension axis of the center bar extends at least substantially perpendicular to the drive axis and/or to the maximum transverse extent of the center bar. In particular, the main extension axis of the center bar extends at least substantially parallel to the main extension plane of the bow-shaped handle, to the main extension plane

of the handle housing, to the separating plane, and/or to the tool holder surface. Preferably, the main extension axis of the center bar extends along the main extension plane of the bow-shaped handle, along the main extension plane of the handle housing, and/or along the separating plane.

The center bar extends, preferably at least in the direction of the main extension axis of the center bar, in particular over a large portion of the surface, preferably over at least 50%, preferably over at least 75% and particularly preferably over 90% of the surface. The upper side, preferably the surface, is arranged facing away from the tool holder. The center bar has a curvature, preferably a curvature adapted to the surface, particularly in the longitudinal direction, preferably in the direction of the main extension axis of the center bar. Alternatively, however, it is also conceivable that the center bar is designed without a curvature in the longitudinal direction.

In one aspect of the invention, which can be considered in particular on its own as well as in conjunction with other aspects of the invention, the invention is based on a handheld power tool system with a sander, in particular the sander specified hereinabove, preferably an orbital sander or a random orbital sander, and with a battery pack for supplying power to the sander. It is proposed that the handheld power tool system comprises a support unit which is designed for placing and/or storing the sander on a base in at least one placement and/or storage position, whereby the battery pack comprises a support surface of the support unit for the at least one placement and/or storage position.

The battery pack can be used to ensure that the sander stands securely. An operator can conveniently park the sander. An operator can park or set down the sander via the battery pack for cleaning, maintenance or the like of the handheld power tool and/or a tool arranged on the sander. A particularly high level of operating convenience can be achieved.

It is conceivable that a support point or a support line forms the support surface. It is also conceivable that the support surface comprises multiple support points, in particular at a distance from one another. Preferably, the support surface is a continuous surface that rests on the base, particularly in the at least one placement and/or storage position. The base can be a floor, a table, or the like. In particular, the battery pack comprises a battery pack housing. The battery pack housing comprises the contact surface.

It is conceivable that the support unit is designed for placing and/or storing the sander on the work surface in at least one further placement and/or storage position. For example, the battery pack comprises a further support surface of the support unit for the at least one further placement and/or storage position. It is also conceivable that the housing unit, in particular the handle housing and/or the drive housing, comprises at least one additional support surface for the placement and/or storage position and/or for the at least one further placement and/or storage position.

In one aspect of the invention, which can be considered in particular on its own as well as in conjunction with other aspects of the invention, the invention is based on a protective device, in particular a wall protection ring, for collision protection of a tool, in particular the tool(s) specified hereinabove, which can be arranged on the handheld power tool and/or a tool holder, in particular a sanding pad, of the handheld power tool with objects in a working environment of the handheld power tool. It is proposed that the protective device comprises a shutter unit and/or a light guide unit for at least one light source.

Particularly efficient and/or precise illumination of a workpiece to be processed by the handheld power tool can be achieved in an advantageous manner. Such a design of the handheld power tool device can counteract blinding of the operator. Advantageously, a particularly high level of operating comfort and/or safety can be achieved.

The protective device is designed in particular as a protective ring. The protective device is preferably intended as a bumper for the handheld power tool. In particular, the protective device is intended to counteract an impact of the tool holder and/or the tool arranged on the tool holder against walls or the like in the working environment. The protective device is preferably arranged on the drive housing in at least one operating state. Preferably, the protective device is intended for arrangement on the fan housing.

The shutter unit is in particular intended to shield an region from the incidence of light from the at least one light source. The shutter unit comprises, e.g., at least one shutter element. Alternatively, however, it is also conceivable that the shutter unit comprises multiple shutter elements. The shutter element is preferably an opaque component. It is conceivable that the shutter element is designed as a reflector, e.g. as a mirror or the like.

The protective device preferably comprises a base body. It is conceivable that the shutter unit, in particular the at least one shutter element, is designed to be integral with the base body. It is conceivable that the shutter unit, in particular the at least one shutter element, is formed by the base body of the protective device. Alternatively, it is also conceivable that the shutter unit, in particular the at least one shutter element, is designed to be separate from the base body. In particular, it is conceivable that the shutter unit, preferably the at least one shutter element, can be attached to the base body, preferably detachably. In this context, the term “detachable” is in particular understood to mean “non-destructively separable”. It is also conceivable that the shutter unit, in particular the at least one shutter element, can be arranged adjustably on the base body, preferably in order to adapt a shielding region by the shutter unit.

The light guide unit is preferably arranged on the base body and/or the shutter unit. For example, the light guide unit comprises at least one light guide element and preferably multiple light guide elements. In particular, the light guide unit is intended to guide and/or emit light from the at least one light source. The at least one light guide element is, e.g., designed as a transparent fiber, tube, or rod. It is also conceivable that the light guide unit comprises at least one emitting element, in particular a light-scattering element, e.g. in the form of a roughening, a scattering glass, plastic or ceramic plate, a scattering plastic film, or the like. The radiating element is preferably arranged on the base body and/or the shutter unit. It is also conceivable that the radiating element is a portion of the light guide element or forms the light guide element.

The at least one light source is, e.g., a portion of the handheld power tool, in particular the sander. Alternatively or additionally, however, it is also conceivable that the light source is a portion of the protective device. The light source is, e.g., designed as an LED, an incandescent lamp, or another light source that a skilled person considers useful.

In one aspect of the invention, which can in particular be considered independently of further aspects of the invention, it is proposed that a center of gravity of the sander is arranged in the handle recess of the bow-shaped handle. A particularly high level of comfort can be achieved when handling handheld power tools. A particularly favorable weight distribution can be achieved in an advantageous

manner. Particularly favorable ergonomic properties can be achieved. Preferably, the center of gravity is arranged within the handle recess at least in a direction perpendicular to the main extension plane of the handle housing, in particular to the main extension plane of the handle housing, when viewed from within the handle recess. A minimum distance of the center of gravity from the separating plane of the handle housing, from the main extension plane of the handle housing and/or from the main extension plane of the bow-shaped handle is preferably smaller than a maximum transverse extent of the handle housing, in particular of the bow-shaped handle, starting from the separating plane of the handle housing, the main extension plane of the handle housing, or the main extension plane of the bow-shaped handle. It is also conceivable that the center of gravity is arranged in the separating plane of the handle housing, in the main extension plane of the handle housing, and/or in the main extension plane of the bow-shaped handle. The maximum transverse extent of the handle housing, in particular of the bow-shaped handle, preferably starting from the separating plane of the handle housing, the main extension plane of the handle housing, and/or the main extension plane of the bow-shaped handle, preferably extends at least substantially perpendicular to the main extension plane of the bow-shaped handle, in particular to the main extension plane of the handle housing, the separating plane, and/or to the drive axis. The maximum transverse extent of the handle housing, in particular of the bow-shaped handle, preferably starting from the separating plane of the handle housing, the main extension plane of the handle housing, and/or the main extension plane of the bow-shaped handle, preferably extends at least substantially parallel to the tool holder surface.

It is further proposed that a ratio of a maximum distance of the tool holder from the center of gravity in the direction of the drive axis to a minimum distance between the drive axis and the center of gravity is at least 2.8 and at most 3.5. Advantageously, a particularly favorable weight distribution can be achieved with regard to handling and/or guiding the handheld power tool. A particularly high level of operating comfort and/or particularly favorable ergonomic properties of the sander can be achieved in an advantageous manner. Preferably, the ratio of the maximum distance of the tool holder from the center of gravity in the direction of the drive axis to the minimum distance between the drive axis and the center of gravity is at least 3.0, particularly preferably at least 3.1. Preferably, the ratio of the maximum distance of the tool holder from the center of gravity in the direction of the drive axis to the minimum distance between the drive axis and the center of gravity at most 3.2. The maximum distance between the tool holder and the center of gravity in the direction of the drive axis is defined in particular by a maximum distance between the tool holder surface and the center of gravity in the direction of the drive axis. The maximum distance between the tool holder and the center of gravity in the direction of the drive axis is in particular at least 30 mm, preferably at least 33 mm, preferably at least 35 mm. The maximum distance between the tool holder and the center of gravity in the direction of the drive axis is in particular at most 50 mm, preferably at most 45 mm, and particularly preferably at most 40 mm. For example, the maximum distance between the tool holder and the center of gravity in the direction of the drive axis is at least substantially 38 mm. Alternatively, however, it is also conceivable that the maximum distance between the tool holder and the center of gravity in the direction of the drive axis is less than 30 mm or greater than 50 mm. The minimum distance

between the drive axis and the center of gravity is at least 8 mm, preferably at least 10 mm. The minimum distance between the drive axis and the center of gravity is at most 20 mm, preferably at most 15 mm. For example, the minimum distance of the drive axis to the center of gravity is at least substantially 12 mm. Alternatively, however, it is also contemplated that the minimum distance of the drive axis to the center of gravity is less than 8 mm or greater than 20 mm. In particular, the drive axis extends at least substantially perpendicular to the tool holder surface. The drive axis extends in particular at least substantially parallel to the main extension plane of the handle housing, in particular to the main extension plane of the bow-shaped handle, and/or to the separation plane. It is conceivable that the drive axis extends in the main extension plane of the handle housing, in the main extension plane of the bow-shaped handle, and/or in the separating plane. The minimum distance between the drive axis and the center of gravity is at least substantially perpendicular to the drive axis. The minimum distance between the drive axis and the center of gravity extends in particular at least substantially parallel to the tool holder surface, to the main extension plane of the handle housing, to the main extension plane of the bow-shaped handle, and/or to the separating plane.

It is also proposed that a maximum height of the sander in the direction of the drive axis has a ratio of at least 2.8 and at most 3.2 to a maximum distance between the tool holder and the center of gravity in the direction of the drive axis, in particular the maximum distance specified hereinabove. The center of gravity of the handheld power tool is particularly low in relation to the maximum height of the sander. Advantageously, a particularly favorable weight distribution and/or particularly favorable ergonomic properties can be achieved with regard to handling and/or guiding the handheld power tool. A particularly high level of operating convenience can be achieved. A particularly high stability of the sander can be achieved in an advantageous manner. The ratio of the maximum height of the sander in the direction of the drive axis to the maximum distance between the tool holder and the center of gravity in the direction of the drive axis is preferably at least 3.0. The ratio of the maximum height of the sander in the direction of the drive axis to the maximum distance between the tool holder and the center of gravity in the direction of the drive axis is preferably 3.1 or less. The maximum height of the sander in the direction of the drive axis preferably extends from the tool holder surface parallel to the drive axis to an outermost point of the handle housing. The maximum height of the sander is preferably at least 100 mm, preferably at least 110 mm. The maximum height of the sander is preferably at most 130 mm, preferably at most 120 mm. Alternatively, however, it is also conceivable that the maximum height of the sander towards the drive axis is less than 110 mm or greater than 130 mm. A ratio of the maximum height of the sander to a maximum height of the handle recess in a direction parallel to the drive axis is preferably at least 4.0, preferably at least 4.3, particularly preferably at least 4.4, and at most 5.0, preferably at most 4.7, particularly preferably at most 4.6. The maximum height of the handle recess is preferably between 20 mm and 30 mm, preferably between 24 mm and 28 mm and particularly preferably at least substantially 26 mm.

It is further proposed that the handle recess of the bow-shaped handle is closed, preferably substantially oval closed. The fingers of the operator can be protected particularly efficiently in an advantageous manner. Advantageously, the handle housing can be additionally supported by the closed bow-shaped handle. A particularly high stability of the

handheld power tool can be achieved in an advantageous manner. In particular, the handle recess is closed when viewed in the main extension plane of the bow-shaped handle, in the main extension plane of the handle housing, and/or in the separating plane. The handle recess is closed in particular by walls of the drive housing, in particular the fan housing of the drive housing, and the handle housing. Alternatively, it is conceivable that the handle recess is completely closed by the walls of the handle housing. The handle recess preferably has an oval shape, in particular when viewed in a direction perpendicular to the main extension plane of the bow-shaped handle, preferably to the main extension plane of the handle housing. Alternatively, however, it is also conceivable that the handle recess has a rectangular shape, a square shape, a circular shape, or any other shape that a skilled person considers useful, in particular when viewed in the direction perpendicular to the main extension plane of the bow-shaped handle, preferably to the main extension plane of the handle housing.

It is further proposed that the battery pack has a battery pack center of gravity which, in a state arranged on the sander, is offset from the main extension plane of the bow-shaped handle. Design-related shifts in the center of gravity of the sander in relation to the main extension plane of the bow-shaped handle can be compensated for. A handheld power tool system with a particularly favorable weight distribution can be provided. A particularly ergonomic handheld power tool system can be provided in an advantageous manner. A particularly high level of operating convenience can be achieved. A minimum distance of the battery pack center of gravity from the separating plane of the handle housing, from the main extension plane of the handle housing, and/or from the main extension plane of the bow-shaped handle is preferably smaller than the maximum transverse extent of the handle housing, in particular of the bow-shaped handle, starting from the separating plane of the handle housing, the main extension plane of the handle housing, or the main extension plane of the bow-shaped handle. In particular, the minimum distance of the battery pack center of gravity from the separating plane of the handle housing, from the main extension plane of the handle housing, and/or from the main extension plane of the bow-shaped handle is less than 20 mm, preferably less than 15 mm, and preferably less than 10 mm.

It is also proposed that the sander comprises a dust outlet, in particular the dust outlet specified hereinabove, whereby the battery pack center of gravity and the dust outlet are arranged on different sides of the main extension plane of the bow-shaped handle. Advantageously, an eccentric displacement of the center of gravity with respect to the main extension plane of the bow-shaped handle can be minimized by the battery pack or the dust outlet and compensated for in a particularly advantageous manner. Advantageously, a sander with a particularly high level of handling comfort can be provided. For example, the center of gravity of the sander and the center of gravity of the battery pack are located on different sides of the main extension plane of the bow-shaped handle. The dust outlet is located in particular on the drive housing, preferably on the fan housing. The dust outlet is arranged in particular on one side of the separating plane, the main extension plane of the bow-shaped handle, and/or the main extension plane of the handle housing. Preferably, the handheld power tool system comprises a dust collection container, e.g. a dust box, a dust bag, or the like. An overall center of gravity of the handheld power tool system relates in particular to the handheld power tool, preferably the sander comprising the battery pack and preferably has no

dust collection container and/or protective device. Preferably, the dust collection container can be attached to the dust outlet, in particular detachably, e.g. by means of a screw connection, a latching connection, or the like. Alternatively, it is also conceivable that at least one dust outlet connecting piece of the dust collection container is designed to be integral with the dust outlet. Preferably, the dust collection container is rotatably mounted on the dust outlet. Alternatively, however, it is also conceivable that the dust collection container can be connected to the dust outlet in a rotationally fixed manner. The dust collection container is in particular intended to collect dust and/or debris, preferably extracted by the extraction fan. Alternatively or additionally, it is conceivable that the dust outlet can be coupled to an active extraction device, e.g. a vacuum cleaner, particularly in the case of a handheld power tool that is designed without an extraction fan. It is also conceivable that the dust collection container can be compressed by contact with the work surface when the handheld power tool is put in the placement and/or storage position, in particular to enable the handheld power tool to be put in the placement and/or storage position. Alternatively or additionally, it is also conceivable that the dust collection container can be displaced in the direction of the dust outlet, in particular displaced relative to the dust outlet via contact with the work surface when the handheld power tool is in the placement and/or storage position.

It is further proposed that a ratio of a minimum distance between the center of gravity of the battery pack in a state arranged on the sander and the drive axis of the drive unit to the maximum longitudinal extent of the tool holder is at least 0.85 and at most 1.2. Advantageously, a particularly favorable weight distribution can be achieved with regard to handling and/or guiding the handheld power tool. A particularly high level of operating convenience can be achieved. The minimum distance between the center of gravity of the battery pack in a state arranged on the sander and the drive axis extends in particular at least substantially parallel to the tool holder surface, the main extension plane of the bow-shaped handle, the main extension plane of the handle housing, and/or the separating plane. The minimum distance between the center of gravity of the battery pack in a state arranged on the sander and the drive axis is at least substantially perpendicular to the drive axis. For example, the minimum distance between the center of gravity of the battery pack and the drive axis is at least substantially 136 mm for a battery pack with 4 Ah electrical charge capacity, at least substantially 148 mm for a battery pack with 8 Ah electrical charge capacity and at least substantially 160 mm for a battery pack with 12 Ah electrical charge capacity. Alternatively, however, other values for the minimum distance between the center of gravity of the battery pack and the drive axis are also conceivable, in particular depending on the weight of the battery pack.

It is further proposed that an overall center of gravity of the handheld power tool system, in particular the one specified hereinabove, is arranged in the handle recess of the bow-shaped handle. A particularly high level of comfort can be achieved when handling handheld power tools. A particularly favorable weight distribution can be achieved in an advantageous manner. A minimum distance of the overall center of gravity from the separating plane of the handle housing, from the main extension plane of the handle housing, and/or from the main extension plane of the bow-shaped handle is preferably smaller than the maximum transverse extent of the handle housing, in particular of the bow-shaped handle, starting from the separating plane of the

handle housing, the main extension plane of the handle housing, and/or the main extension plane of the bow-shaped handle. Preferably, the overall center of gravity is arranged in the separating plane of the handle housing, in the main extension plane of the handle housing, and/or in the main extension plane of the bow-shaped handle. Preferably, the overall center of gravity of the handheld power tool system is arranged at least with a battery pack with a single-layer battery cell arrangement within the handle recess of the bow-shaped handle. Preferably, the position of the overall center of gravity of the handheld power tool system with a battery pack with a single-layer battery cell arrangement is such that the handheld power tool can be placed on the tool holder, in particular without tilting.

A minimum distance between the drive axis and the overall center of gravity has a value of at least 55 mm to at least 70 mm, in particular depending on the weight of the battery pack. The minimum distance between the drive axis and the overall center of gravity has a value of at most 65 mm to at most 75 mm, depending in particular on the weight of the battery pack. A maximum distance of the tool holder to the overall center of gravity in the direction of the drive axis has a value of at least 45 mm to at least 50 mm, in particular depending on the weight of the battery pack. The maximum distance between the tool holder and the overall center of gravity in the direction of the drive axis has a value of at most 52 mm to at most 55 mm, depending in particular on the weight of the battery pack.

It is further proposed that a ratio of the minimum distance between the center of gravity of the battery pack in a state arranged on the sander and the drive axis of the drive unit to the maximum longitudinal extent of the handle recess of the bow-shaped handle is at least 2.0 and at most 3.0. Advantageously, particularly precise and/or controlled handling of the sander can be achieved in an advantageous manner. Particularly safe and/or precise working with the sander can be achieved in an advantageous manner. A handheld power tool system with particularly favorable ergonomic properties can be provided in an advantageous manner.

It is also proposed that the handle housing at least partially covers the center bar on both sides in directions perpendicular to the main extension plane of the handle housing. A particularly large gripping surface can be provided for the operator in an advantageous manner. The operator can be prevented from coming into contact with the drive housing when using the handheld power tool device. A particularly high level of operating convenience can be achieved in an advantageous manner. The handle housing covers the center bar on both sides in the directions perpendicular to the main extension plane of the handle housing, in particular at a height of at least 10%, preferably at least 20%, and preferably at least 30%. The center bar is not covered by the handle housing, particularly in the longitudinal direction. It is also conceivable that the handle housing largely covers the center bar on both sides at a height in the directions perpendicular to the main extension plane of the handle housing, in particular at least 50%, preferably at least 70%, and preferably at least 90%.

It is also proposed that an end of the handle housing facing the center bar, preferably the end already specified hereinabove, in particular an end surrounding the drive housing in a substantially ring-like manner, is arranged at a distance from the drive housing, in particular at a radial and axial distance, preferably with respect to the drive axis. A particularly large grip region can be made provided to the operator in an advantageous manner. The operator can be prevented from coming into contact with the drive housing

when using the handheld power tool device. A particularly high level of operating convenience can be achieved in an advantageous manner. Preferably, the end is arranged at a distance from the surface. Preferably, the end surrounds the motor housing of the drive housing like a ring. In particular, the motor housing is arranged on the center bar. The motor housing is preferably arranged on a side of the center bar facing away from the tool holder, in particular above the center bar with respect to the tool holder. The motor housing is preferably enclosed by the handle housing in a plane perpendicular to the drive axis. Preferably, the motor housing is at least substantially completely enclosed by the handle housing in the radial direction starting from the drive axis. The phrase "at least substantially complete" can be understood at least 50%, preferably at least 75%, and particularly preferably at least 90% of a total volume and/or a total mass of an object. Preferably, the motor housing is designed to be at least partially integral with the center bar. The term "at least one unit or object and at least one further unit or object are designed to be at least partially integral with one another" is in particular understood to mean that at least one element of the unit or object is designed to be integral with at least one further element of the further unit or object. The term "integral" can be understood to mean at least connected in a bonded manner, e.g. by a welding process, a soldering process, a bonding process, an injection molding process, and/or another process that a skilled person considers useful, and/or advantageously designed in one piece, e.g. by production from a casting and/or by production in a single or multi-component injection molding process and advantageously from a single blank. The handle housing, in particular the end of the handle housing, is preferably arranged at a distance from the drive housing, in particular the motor housing.

It is further proposed that the end of the handle housing facing the center bar is adapted to the shape of the drive housing, in particular the center bar and/or the surface. A particularly large gripping surface for an operator can be achieved in an advantageous manner, along with a particularly small space requirement. Preferably, said end comprises two recesses for the center bar. It is conceivable that the end is at a constant distance from the drive housing in the direction of the drive axis. Alternatively, however, it is also conceivable that the distance between the end and the drive housing varies in the direction of the drive axis.

It is also proposed that the center bar comprises a holder for a user interface and/or a communication module. A particularly space-saving arrangement of a user interface can be achieved in an advantageous manner. The functionality of the handheld power tool device can be extended to save space in an advantageous manner. Particularly protected and secure positioning of a user interface can be achieved in an advantageous manner. The holder is preferably designed as a recess in the center bar. The user interface comprises, e.g., one or multiple input elements. The input element of the user interface can, for example, be designed as a button, a rotary wheel, a switch, or the like. For example, the handheld power tool can be switched on and/or off via the user interface, in particular via the at least one input element of the user interface, and/or a rotational speed of the handheld power tool, preferably the sander, can be adjusted or the like. It is conceivable that the user interface comprises one or multiple output elements, e.g. a display, a light element, in particular an LED, or the like. For example, the user interface, in particular the at least one output element, can be used to output the rotational speed of the handheld power tool, preferably the sander, a charge status of the battery

pack, a device status of the handheld power tool, or the like. Preferably, the user interface is connected via data technology to a control unit of the handheld power tool, in particular the sander, preferably by in a wired and/or wireless manner. The control unit is intended in particular for controlling the drive unit. In particular, the control unit comprises at least one processor and one memory element, as well as an operating program stored on the memory element. The memory element is preferably designed as a digital storage medium, e.g. a hard disk or the like. The holder is preferably arranged on a side of the motor housing facing away from the bow-shaped handle. In particular, the holder is arranged on a side of the handle housing facing away from the battery pack interface of the handheld power tool. In particular, the communication module is configured to communicate with another device, e.g. a smartphone, a laptop, a server, a cloud, a gateway, or the like. Preferably, the communication module is connected to the control unit and/or the user interface using data technology, in particular in a wireless and/or by wired manner. It is conceivable that the communication module is a portion of the user interface. Alternatively, however, it is also conceivable that the communication module is designed and/or arranged to be separate from the user interface. For example, it is conceivable that the communication module is arranged in the handle housing, in particular in the bow-shaped handle. It is also conceivable that the communication module is a portion of the control unit. The communication module can comprise one or multiple data transmission technologies, e.g. Bluetooth, NFC, or the like. The control unit is preferably located in the handle housing.

In one aspect of the invention, which can in particular be considered independently of further aspects of the invention, it is proposed that the drive housing comprises at least one air outlet opening which is at least partially covered by the handle housing, in particular when viewed at least in a main outlet direction of the air outlet opening. A particularly favorable exhaust air flow can be achieved in an advantageous manner. Advantageously, the operator can be protected from exhaust air particularly efficiently and/or in a structurally simple manner. A particularly high level of operating convenience can be achieved in an advantageous manner. Preferably, the air outlet opening is at least partially covered by the handle housing, at least when viewed in a plane perpendicular to the drive axis. It is conceivable that the handle housing covers at least most of the air outlet opening, preferably at least 50%, preferably at least 75% and particularly preferably at least 90%, in particular when viewed in a plane perpendicular to the drive axis. It is also conceivable that the handle housing completely covers the air outlet opening, particularly when viewed in a plane perpendicular to the drive axis. The main outlet direction is determined by the arrangement and/or geometry of the air outlet opening. For example, the main outlet direction extends at least substantially perpendicular to the drive axis and/or at an angle to the drive axis, preferably inclined in the direction of the tool holder. The air outlet opening is preferably covered by the end of the handle housing, in particular when viewed in a plane perpendicular to the drive axis. Preferably, the air outlet opening is arranged above the center bar, preferably in relation to the tool holder and/or the surface.

It is further proposed that the drive housing comprises a motor housing, in particular the motor housing specified hereinabove, which is arranged on the center bar and comprises the air outlet opening. The hot engine exhaust air can be efficiently dissipated while at the same time providing

particularly efficient protection for the operator from the engine exhaust air. A handheld power tool device with a particularly high level of operating convenience can be provided. The motor housing preferably comprises a plurality of air outlet openings. The air outlet openings are, e.g., arranged at a distance from each other in a circumferential direction, which in particular extends in a plane perpendicular to the drive axis.

It is also proposed that an opening is formed between the end of the handle housing facing the center bar and the drive housing to allow air to escape from the air outlet opening. The hot engine exhaust air can be efficiently dissipated while at the same time providing particularly efficient protection for the operator from the engine exhaust air in an advantageous manner. A handheld power tool device with a particularly high level of operating convenience can be provided. The opening is preferably arranged opposite the upper side. The opening is in particular formed by arranging the end of the handle housing at a distance from the drive housing, preferably the motor housing.

In one aspect of the invention, which can in particular be considered independently of further aspects of the invention, it is proposed that the drive housing comprises a closed cable channel. Efficient protection for a cable connection can be advantageously provided. A particularly space-saving and/or protected arrangement of a cable connection can be achieved in an advantageous manner. It is conceivable that at least a part, in particular a large part, preferably at least 50%, preferably at least 70%, of the cable channel is enclosed by the handle housing, in particular when viewed in a plane perpendicular to the drive axis. An open end of the cable channel is preferably arranged on the holder. A further open end of the cable channel is preferably arranged in an inner region of the handle housing. In particular, the cable channel extends from the holder to the inside of the handle housing. The cable channel extends over the entire motor housing, for example, particularly in the direction of the drive axis. Preferably, the handheld power tool, in particular the sander and/or the handheld power tool device comprises an electrical connection unit. In particular, the electrical connection unit comprises at least one electrical connection element, e.g. a cable. The electrical connection unit, in particular the electrical connection element, is preferably intended to connect the user interface to the control unit, in particular at least in terms of data technology and/or control technology. Preferably, the user interface can be supplied with electrical energy via the electrical connection unit, preferably via the at least one electrical connection element and/or a further electrical connection element of the electrical connection unit. The electrical connection unit, in particular the at least one electrical connection element and/or the further electrical connection element, is/are arranged in the cable channel. The holder preferably comprises at least one electrical connection for the user interface and/or the communication interface. At least the electrical connection element and/or the further electrical connection element is/are connected to the connection.

It is also proposed that the drive housing comprises at least two housing shells, between which the cable channel is arranged. Advantageously, it is particularly easy to install a cable connection in the cable channel. Advantageously, a closed cable channel can be provided in a simple design and, in particular, with a particularly high level of convenience when installing a cable connection. In particular, the drive housing has a shell design, with the two housing shells acting as half shells. The two housing shells of the drive housing are preferably connected to each other in a drive

housing separating plane of the drive housing. The drive housing separating plane preferably extends at least substantially parallel to the separating plane of the handle housing. It is conceivable that the drive housing separating plane corresponds to the separating plane of the handle housing. Alternatively, however, it is also conceivable that the drive housing separating plane extends at an angle to the separating plane. Preferably, the cable channel extends inside the outer walls of the housing shells, in particular inside at least one outer wall of one of the housing shells. Preferably, the cable channel connects to a contact surface of the outer walls of the two housing shells.

It is also proposed that one housing shell of the housing shells forms a groove for the cable channel and another housing shell of the housing shells forms a cover for the cable channel. A closed cable channel with a simple design can be provided in a structurally advantageous manner.

It is further proposed that the cable channel is arranged at least partially on, in particular in, the center bar. A closed cable channel can be achieved in a particularly space-saving and advantageous manner. It is conceivable that at least a portion of the cable channel extends through the center bar, e.g. at least 5%, 10%, or 20% of the cable channel.

Further proposed in particular is a handheld power tool, in particular the handheld sander specified hereinabove, preferably the sander specified hereinabove comprising the handheld power tool device. A handheld power tool with a particularly low overall height can be provided in an advantageous manner. A handheld power tool with particularly favorable handling characteristics, in particular a particularly large gripping surface for the operator, can be provided.

It is also proposed that the sander comprises a tool holder, in particular the tool holder specified hereinabove for holding a tool, in particular the tool specified hereinabove, whereby the tool holder and/or the tool are/is at a distance from the work surface in the at least one placement and/or storage position. Advantageously, an excessive load on the tool holder and/or the tool in the at least one placement and/or storage position of the sander can be counteracted. Damage to the tool holder and/or the tool, in particular by the work surface, can be counteracted. It is possible to conveniently replace the tool. A particularly high level of operating convenience can be achieved, in particular when replacing tools and/or performing maintenance work or the like on the tool holder.

It is further proposed that the sander can be placed in the at least one placement and/or storage position solely via the support surface. It is advantageous to counteract the formation of signs of wear caused by contact between the sander, in particular the housing unit, and the work surface. The sander can be placed particularly gently in an advantageous manner. Preferably, the sander is at a distance from the work surface in the at least one placement and/or storage position. In particular, the sander is connected to the work surface via the battery pack in the at least one placement and/or storage position, and can preferably be placed on the work surface. In the at least one placement and/or storage position, only the contact surface on the battery pack is in contact with the work surface.

It is further proposed that a main extension plane of the supporting surface extends at an angle other than 90° to a main extension plane of the tool holder surface of the tool holder of the sander. Advantageously, a particularly favorable weight distribution can be achieved when placing the sander down. Particularly safe placement of the sander can be achieved in an advantageous manner. By tilting the tool holder surface in this way, the tool can be mounted and

removed particularly conveniently in the placement and/or storage position of the sander. A particularly high level of user comfort can be achieved in an advantageous manner. In particular, a main extension axis of the handheld power tool, in particular the sander, extends at an angle other than 90° to the main extension plane of the support surface. It is conceivable that the tool holder surface extends at least substantially parallel to the main extension axis of the handheld power tool, in particular the sander. In particular, a handle axis of the handle housing extends at an angle other than 90° to the main extension plane of the support surface in the placement and/or storage position. The handle axis is defined in particular by the bow-shaped handle. The handle axis preferably extends in the separating plane and/or in the main extension plane of the handle housing, in particular in the main extension plane of the bow-shaped handle. The handle axis corresponds in particular to a main extension axis of a handle tube of the bow-shaped handle.

It is also proposed that the main extension plane of the tool holder surface and the main extension plane of the supporting surface include an angle which is greater than 80°. Advantageously, the sander can be placed in a particularly safe and space-saving manner. Preferably, the angle enclosed by the main extension plane of the tool holder surface and the main extension plane of the supporting surface has a value of less than 90° and greater than 80°. In particular, the tool holder, preferably the tool holder surface, is inclined in a direction away from the work surface in the placement and/or storage position. Particularly preferably, the angle enclosed by the main extension plane of the tool holder surface and the main extension plane of the supporting surface has a value of at least substantially 86°.

It is further proposed that a normal of the main extension plane of the support surface intersects the battery pack and a center of gravity of the sander. Advantageously, the sander can be placed in a particularly stable manner. Preferably, the normal of the main extension plane of the support surface intersects a geometric center of the support surface and/or the battery pack center of gravity, in particular at least a nearby region of the geometric center of the support surface and/or a nearby region of the battery pack center of gravity. In this context, the term "nearby region" is in particular context understood to mean a range that is at a maximum distance from a reference point that is preferably less than 7%, preferably less than 5%, of a maximum longitudinal extent of the battery pack.

It is also proposed that one edge of the battery pack housing of the battery pack forms the contact surface. Advantageously, the sander can be placed on the base in a particularly stable manner. Preferably, an edge of the battery pack housing facing away from the tool holder forms the contact surface. Preferably, the support surface and a support point of the sander span a support plane of a, in particular further placement and/or storage position of the support unit.

It is also proposed that a main extension axis of the battery pack extends at least substantially perpendicular to the main extension axis of the sander. Advantageously, the sander can be placed on the work surface in a particularly stable manner. Preferably, the edge of the battery pack housing extends at least substantially perpendicular to the main extension axis of the handheld power tool.

It is further proposed that the sander comprises only one support point, in particular the one specified hereinabove, which comes to rest in a support plane, in particular the one specified hereinabove, of an in particular further placement and/or storage position of the support unit, in particular the one specified hereinabove. Advantageously, wear on the

sander, in particular the housing unit, can be minimized during contact with the work surface, in particular at a quite high level of stability during placement of the sander. The support point is preferably located on the handle housing or the drive housing. The contact point is preferably located on a surface of the palm grip. The palm grip preferably has a round shape. The surface of the palm grip, on which the support point is arranged, is preferably curved. Alternatively, however, it is also conceivable that the surface of the palm grip on which the support point is arranged is flat. The support point of the sander is preferably arranged on a side of the handle housing facing away from the tool holder. Alternatively, however, it is also conceivable that at least one support elevation is arranged on the palm grip, which comes to rest in the support plane of the at least one, in particular further placement and/or storage position. The at least one support elevation is intended in particular to form a support line for placement of the handheld power tool. The support elevation is, e.g., designed as an elongated bar whose main extension axis extends at least substantially perpendicular to the main extension axis of the handheld power tool, to the separating plane, to the main extension plane of the bow-shaped handle, and/or to the main extension plane of the handle housing. Furthermore, it is also conceivable that the palm grip comprises at least two support elevations, preferably in a direction perpendicular to the separating plane, which are at a distance from each other and designed as humps or the like and which are preferably situated in the support plane of the at least one, in particular further placement and/or storage position. Given a design having at least one raised support on the palm grip, it is feasible to place the handheld power tool without the battery pack in an additional placement and/or storage position, whereby the handheld power tool can be placed on the work surface via the at least one raised support on the palm grip and a further support point on the handle housing, in particular in a region of the battery pack interface. Additionally or alternatively, it is also feasible for at least one raised support to be arranged on the handle housing, in particular in the region of the battery pack interface.

It is further proposed that the support unit is designed for placing and/or storing the sander on the work surface in at least one, in particular the further placement and/or storage position specified hereinabove, whereby the battery pack comprises a further support surface of the support unit for the further placement and/or storage position. Advantageously, a particularly high degree of flexibility can be achieved while safely placing the sander on a work surface. For example, the edge of the battery pack forms the additional support surface of the support unit for the additional placement and/or storage position. In the further placement and/or storage position, the tool holder surface preferably extends at least substantially parallel to a support plane of the further placement and/or storage position. In the further placement and/or storage position, the drive axis extends in particular at least substantially perpendicular to the support plane of the further placement and/or storage position. In the further placement and/or storage position, the further support surface and the support point of the sander in particular come to lie in the support plane of the further placement and/or storage position. The handheld power tool system is preferably in contact with the work surface in the further placement and/or storage position via the further support surface on the battery pack and the support point on the sander.

It is further proposed that the shutter unit is intended to block a light emission from at least one light source, at least

in the direction of an intended operator side. Advantageously, an operator can be prevented from being blinded when working with the safety guard. It can enable particularly safe and/or comfortable working with the protective device. In particular, the shutter unit is intended to block the light emission of the at least one light source in a direction away from the tool holder. The operator side corresponds in particular to a position of the operator in relation to the handheld power tool, in particular the light source, which the operator assumes when working on a workpiece.

In one aspect of the invention, which can in particular be considered independently of further aspects of the invention, it is proposed that the protective device comprises a light source unit. A workpiece can be illuminated with particular precision during processing in an advantageous manner. Advantageously, the light source unit is particularly easy to maintain and/or replace. A particularly high level of operating convenience can be achieved. The light source unit preferably comprises at least one light source, e.g. the light source or multiple light sources specified hereinabove. The light source unit, in particular the at least one light source, is preferably arranged on the base body, the shutter unit, or the light guide unit.

In one aspect of the invention, which can in particular be considered independently of further aspects of the invention, it is proposed that the protective device comprises at least one energy source and/or at least one operating element for at least one light source, in particular the one specified hereinabove. The functionality of the protective device can be advantageously extended. Such a design of the protective device enables particularly comfortable working. The energy source can be in the form of a battery, rechargeable battery, or the like. The energy source is particularly intended for supplying energy to the light source, preferably the light source unit. Preferably, the energy source is connected to the light source, in particular the light source unit, at least in one operating state. Alternatively or additionally, however, it is also conceivable that the light source, preferably the light source unit, can be supplied with electrical energy via the handheld power tool, in particular the sander, e.g. via an electrical plug connection between the protective device and the handheld power tool. The at least one operating element of the protective device is preferably arranged on the base body. The at least one operating element of the protective device can, e.g., be designed as a button, a rotary wheel, a switch, or the like. The at least one light source can, e.g., be switched on and/or off and/or an illumination parameter, such as a light color, intensity or the like, can be adjusted via the at least one control element of the protective device.

In one aspect of the invention, which can in particular be considered independently of further aspects of the invention, it is proposed that the protective device comprises a sensor unit for detecting operation of the handheld power tool, in particular for activating at least one light source, in particular the light source unit specified hereinabove, preferably the light source unit, during operation of the handheld power tool. Advantageously, activation of the at least one light source can be automated in an advantageous manner. A particularly high level of user comfort can be achieved in an advantageous manner. The sensor unit is preferably arranged on the base body, on the shutter unit, or on the light guide unit. The sensor unit preferably comprises at least one sensor element, e.g. a vibration sensor or the like, for detecting operation of the handheld power tool. The sensor unit preferably comprises control electronics for processing the sensor signals. The control electronics are preferably connected to the at least one light source, in particular the light

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source unit, for control purposes. In particular, the control electronics are intended to switch on the light source when operation of the handheld power tool is detected. It is conceivable that the control electronics are a portion of the control unit of the handheld power tool. Alternatively, however, it is also conceivable that the control electronics are designed separately from the control unit.

In one aspect of the invention, which can in particular be considered independently of further aspects of the invention, it is proposed that the protective device is at least partially made of a thermoplastic elastomer, in particular comprising a protective layer consisting of at least partially of a thermoplastic elastomer. A particularly high level of impact protection can be achieved in an advantageous manner. A particularly durable protective device can be provided.

Preferably, the base body of the protective device is at least partially made of a thermoplastic elastomer. Preferably, at least one surface of the protective device facing away from the handheld power tool, in particular the base body, is made of a thermoplastic elastomer. The protective layer can be a cover placed over the base body, a layer glued to the base body, or the like. Alternatively or additionally, it is also conceivable that the protective layer is clamped, screwed, or the like onto the base body. It is also conceivable that the protective layer is injection-molded around the base body, in particular that the protective layer is injection-molded onto the base body, e.g. in a two-component injection molding process.

In one aspect of the invention, which can in particular be considered independently of further aspects of the invention, it is proposed that the protective device comprises a fastening unit for detachable fastening to the handheld power tool with at least two fixing rails which extend parallel to one another and comprise at least two latching recesses. Robust attachment of the protective device to the handheld power tool is easy to implement in an advantageous manner. The handheld power tool, in particular the drive housing, preferably the fan housing, comprises in particular a protective fastening unit corresponding to the fastening unit. The protective fastening unit comprises, e.g., at least one fixing rail. The fixing rail of the protective fastening unit is arranged between the fixing rails of the fastening unit, in particular in a state of the protective device, in particular the base body, arranged on the handheld power tool. The fixing rails of the fastening unit and the fixing rail of the protective fastening unit are preferably intended for fixing in an axial direction relative to one another, preferably with respect to the drive axis. The protective fastening unit preferably comprises two latching elements, in particular latching hooks or the like. In particular, the latching elements are intended to interact with the latching recesses of the fastening unit to fasten the protective device, in particular the base body, to the handheld power tool. Alternatively, it is also conceivable that the design of the fastening unit and the protective fastening unit are interchanged.

In one aspect of the invention, which can in particular be considered independently of further aspects of the invention, it is proposed that the protective device has an asymmetrical basic shape, in particular adapted to a dust outlet of the handheld power tool, in particular the dust outlet specified hereinabove. In particular, a shape of the base body deviates from a circular shape along its longitudinal extent in at least one section, preferably in a region of the dust outlet.

Further proposed is a handheld power tool system, in particular the one specified hereinabove, in particular comprising the handheld power tool and protective device specified hereinabove. Particularly efficient and/or precise illu-

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mination of a workpiece to be processed by the handheld power tool can be achieved. Such a design of the protective device can counteract blinding of the operator. Advantageously, a particularly high level of operating comfort and/or safety can be achieved.

In addition, the invention is based on a method for manufacturing a protective device, in particular the protective device specified hereinabove. It is proposed that the protective device be made at least partially of a thermoplastic elastomer. A particularly durable protective device can be provided in an advantageous manner. Particularly efficient impact protection can be achieved in an advantageous manner. Preferably, the base body is made at least partially of a thermoplastic elastomer, e.g. in a single or multi-component injection molding process.

It is further proposed that the handheld power tool device comprises a tool holder, in particular the tool holder specified hereinabove for holding a tool, in particular the tool specified hereinabove, in particular a sanding means, whereby a maximum distance of the extension arm from the drive axis is greater than a maximum distance of the tool holder from the drive axis, in particular when viewed in the separating plane of the handle housing.

A particularly wide support of the drive housing on the handle housing can be advantageously achieved. Advantageously, a particularly high level of stability can be achieved when guiding the handheld power tool, in particular the handheld power tool device.

Advantageously, particularly precise work can be enabled using the handheld power tool. Preferably, a maximum distance of the portion of the damping unit on the extension arm to the drive axis is greater than the maximum distance of the tool holder to the drive axis, in particular when viewed in the separating plane of the handle housing.

It is further proposed that a ratio of a maximum transverse extent of the portion of the damping unit to a minimum distance of the portion of the damping unit from the handle axis of the handle housing is at least 0.9 and at most 1.05. Advantageously, torques around the handle axis can be supported particularly efficiently. A particularly high level of stability can be achieved in an advantageous manner. Particularly precise work can be achieved by the handheld power tool device. The maximum transverse extent of the portion of the damping unit extends in particular at least substantially perpendicular to the main extension plane of the handle housing, preferably the main extension plane of the bow-shaped handle and/or the separating plane. The maximum transverse extent of the portion of the damping unit preferably extends at least substantially parallel to the tool holder surface. The minimum distance of the portion of the damping unit from the handle axis is preferably measured parallel to the separating plane, in particular to the main extension plane of the bow-shaped handle, preferably the main extension plane of the handle housing. The maximum transverse extent of the portion of the damping unit is preferably at least 40 mm, preferably at least 45 mm. The maximum transverse extent of the portion of the damping unit is preferably no more than 60 mm, preferably no more than 50 mm. The ratio between the maximum transverse extent of the portion of the damping unit and the maximum height of the sander is preferably at most 0.35, preferably at most 0.4. The ratio between the maximum transverse extent of the portion of the damping unit and the maximum height of the sander is preferably at most 0.5, preferably at most 0.45. The minimum distance between the portion of the damping unit and the handle axis is preferably at least 45 mm, preferably at least 50 mm. The minimum distance

between the portion of the damping unit and the handle axis is preferably at most 60 mm, preferably at most 55 mm. Alternatively, however, it is also conceivable that the minimum distance between the portion of the damping unit and the handle axis is less than 45 mm or greater than 60 mm.

It is also proposed that a ratio between the maximum transverse extent of the portion of the damping unit and a minimum distance of the portion of the damping unit from the drive axis is at least 0.7, preferably at least 0.73, and at most 0.8, preferably at most 0.77. Advantageously, tipping forces can be supported particularly efficiently. A particularly stable handheld power tool device can be provided. The minimum distance of the portion of the damping unit to the drive axis is preferably measured parallel to the separating plane, in particular to the main extension plane of the bow-shaped handle, preferably the main extension plane of the handle housing. The minimum distance between the portion of the damping unit and the drive axis is preferably at least 55 mm, preferably at least 60 mm. The minimum distance between the portion of the damping unit and the drive axis is preferably no more than 70 mm, preferably no more than 65 mm. Alternatively, however, it is also conceivable that the minimum distance between the portion of the damping unit and the drive axis is less than 55 mm or greater than 70 mm.

It is further proposed that a ratio of a maximum height of the housing unit to the minimum distance of the portion of the damping unit on the extension arm from the handle axis of the handle housing is at most 2.25. A particularly wide support of the drive housing on the handle housing can be achieved. Advantageously, a particularly high level of stability can be achieved when handling the handheld power tool, in particular the sander. The maximum height of the housing unit preferably extends in the separating plane, in particular in the main extension plane of the handle housing, preferably in the main extension plane of the bow-shaped handle. The maximum height of the housing unit extends parallel to the drive axis in particular. The maximum height of the housing unit is preferably at least 90 mm, preferably at least 100 mm. The maximum height of the housing unit is preferably at most 120 mm, preferably at most 110 mm. Alternatively, however, it is also conceivable that the maximum height of the housing unit in the direction of the drive axis is less than 90 mm or greater than 120 mm. A ratio of a maximum extent of the damping unit, in particular a maximum distance between the damping elements, in the direction of the drive axis to the maximum height of the housing unit is preferably at least 0.8, preferably at least 0.85. A ratio of the maximum extent of the damping unit, in particular a maximum distance between the damping elements, in the direction of the drive axis to the maximum height of the sander is preferably at least 0.7, preferably at least 0.75. The maximum extent of the damping unit, in particular the maximum distance between the damping elements, in the direction of the drive axis is preferably at least 85 mm, preferably at least 90 mm. Alternatively, however, it is also conceivable that the maximum extent of the damping unit, in particular the maximum distance between the damping elements, in the direction of the drive axis is less than 85 mm.

It is also proposed that a ratio of the minimum distance of the portion of the damping unit from the drive axis in a direction perpendicular to the drive axis to a maximum longitudinal extent of the housing unit is greater than 0.3. A particularly wide support of the drive housing on the handle housing can be achieved. Advantageously, a particularly high level of stability can be achieved when handling the

handheld power tool, in particular the sander. The maximum longitudinal extent of the housing unit is preferably perpendicular to the drive axis. The maximum longitudinal extent of the housing unit preferably extends at least substantially parallel to the tool holder surface. The maximum longitudinal extent of the housing unit extends in particular in the separating plane, preferably in the main extension plane of the handle housing, in particular in the main extension plane of the bow-shaped handle. The maximum longitudinal extent of the housing unit is preferably at least 175 mm, preferably at least 180 mm. The maximum longitudinal extent of the housing unit is preferably no more than 190 mm, preferably no more than 185 mm. Alternatively, however, it is also conceivable that the maximum longitudinal extent of the housing unit is less than 175 mm or greater than 190 mm.

It is further proposed that a ratio of the minimum distance between the portion of the damping unit and the drive axis to the minimum distance between the portion of the damping unit and the handle axis of the handle housing is at least 1.2 and at most 1.3. A handheld power tool with a particularly favorable support of the drive housing on the handle housing can be provided.

It is further proposed that the handle housing comprises a bow-shaped handle, in particular the bow-shaped handle specified hereinabove, which is supported by the portion of the damping unit and/or the extension arm on the drive housing to form the closed handle recess of the bow-shaped handle. A particularly efficient support of the drive housing on the handle housing can be achieved along with simultaneous vibration decoupling of the handle housing from the drive housing. A particularly high level of handling stability can be achieved with a particularly high level of operating convenience.

It is also proposed that the portion of the damping unit and/or the extension arm is arranged in the direction of the handle axis at the height of the handle recess. A particularly wide support of the drive housing on the handle housing can be achieved. Advantageously, a handheld power tool device can be provided with particularly high stability and, at the same time, particularly efficient vibration decoupling. Preferably, the portion of the damping unit on the extension arm and/or the extension arms, viewed in the direction of the handle axis, is arranged at least substantially completely within the maximum longitudinal extent of the handle recess.

It is further proposed that the portion of the damping unit arranged on the extension arm comprises at least two damping elements arranged at a distance from each other. A particularly favorable ratio between stable support of the drive housing on the handle housing and vibration decoupling of the handle housing from the drive housing can be achieved in an advantageous manner. The damping elements are preferably arranged on opposite sides of the extension arm. Preferably, the damping elements are arranged on different sides of the separating plane, in particular the main extension plane of the handle housing, preferably the main extension plane of the bow-shaped handle. Preferably, the damping elements are arranged in mirror symmetry with respect to the separating plane, in particular the main extension plane of the handle housing, preferably the main extension plane of the bow-shaped handle. The extension arm preferably comprises two damping element holders for the two damping elements specified hereinabove, or the damping element holder is configured to hold the two damping elements.

It is also proposed that the housing unit comprises a fan housing, in particular the fan housing specified hereinabove for holding an extraction fan, in particular the extraction fan specified hereinabove, whereby the extension arm is arranged on the fan housing. A particularly wide support for the drive housing can be achieved in an advantageous manner. Particularly stable guidance of the handheld power tool, in particular the handheld power tool device, can be achieved with simultaneous vibration decoupling of the drive housing from the handle housing. In particular, the extraction fan is a portion of the handheld power tool, preferably the sander. Alternatively, it is also conceivable that the handheld power tool, in particular the sander, is designed without an extraction fan. Preferably, the extension arm is designed to be at least partially integral with the fan housing. The extension arm is arranged in particular on a side of the fan housing facing the battery pack interface. It is conceivable that the extension arm is, e.g., preferably arranged between two swing legs of the tool holder on a handheld power tool designed as an orbital sander, which comprises a rectangular tool holder. In particular, the swing legs are made of an elastic material, preferably plastic. The swing legs are preferably connected to the drive housing, in particular to the fan housing.

In one aspect of the invention, which can in particular be considered independently of further aspects of the invention, it is proposed that the drive housing comprises a motor housing, in particular the motor housing specified hereinabove for holding the drive unit and a fan housing, in particular the fan housing specified hereinabove for holding an extraction fan, in particular the extraction fan specified hereinabove, whereby the motor housing and the fan housing are designed to be at least partially integral with one another. A tolerance chain can be kept particularly low in an advantageous manner. A particularly high stability of the drive housing can be achieved. The fan housing and the motor housing comprise common housing shells, in particular the housing shells of the drive housing specified hereinabove. The two housing shells each comprise one half of the fan housing and one half of the motor housing.

In one aspect of the invention, which can in particular be considered independently of further aspects of the invention, it is proposed that the drive housing comprises a motor housing, in particular the motor housing specified hereinabove, on which at least one damping element of the damping unit is arranged such that the damping element is only partially enclosed by the handle housing when viewed in a sectional plane perpendicular to the drive axis through the damping element. Advantageously, the handle housing can be supported on the drive housing at a point on the handle housing that is particularly far outwards in the axial direction, in particular in relation to the drive axis. Particularly efficient support can be achieved with simultaneous vibration decoupling. A particularly high level of stability can be achieved when handling the handheld power tool, in particular the sander. Preferably, the cutting plane through the damping element only intersects a portion of the end of the handle housing facing the center bar, in particular due to the two recesses in the end for the center bar.

In one aspect of the invention, which can in particular be considered independently of further aspects of the invention, it is proposed that the damping unit comprises at least one sealing element, in particular a sealing ring, in order to block an inlet of exhaust air into the housing unit, in particular into the handle housing, whereby the sealing element, in particular the sealing ring, is arranged between the handle housing and the drive housing. Warm engine air can in an

advantageous manner be prevented from flowing into the handle housing. Overheating of the handheld power tool, in particular the sander, can be counteracted. It is conceivable that the sealing element is arranged, in particular designed, as a closed sealing ring. The sealing element surrounds the motor housing in a circumferential direction, which in particular extends in a plane perpendicular to the drive axis, preferably at least substantially completely, in particular in an angular range of at least 270°, preferably at least 330°, preferably at least 350°, and particularly preferably completely. The sealing element preferably fills a gap between the motor housing and the handle housing at least substantially completely. The sealing element is preferably made of an elastic material, preferably an elastic plastic. Preferably, the material hardness of the sealing element preferably is different from the material hardness of the damping elements. The material hardness of the sealing element is preferably lower than the material hardness of the damping elements. Alternatively, it is conceivable that the sealing element and the damping elements have an identical material hardness. The sealing element is arranged above at least one air outlet in relation to the tool holder.

The handheld power tool, in particular the sander, the handheld power tool system, the protective device, the method, and/or the handheld power tool device are not intended to be limited to the application and embodiment described hereinabove. In particular, the handheld power tool, in particular the sander, the handheld power tool system, the method, and/or the handheld power tool device can/can comprise a number of individual elements, components, and units as well as method steps that deviates from a number specified herein in order to fulfill a mode of operation described herein. Moreover, regarding the ranges of values indicated in this disclosure, values lying within the limits specified hereinabove are also intended to be considered as disclosed and usable as desired.

BRIEF DESCRIPTION OF THE DRAWINGS

Further advantages follow from the description of the drawings hereinafter. Two exemplary embodiments of the invention are shown in the drawings. The drawings, the description, and the claims contain numerous features in combination. The skilled person will appropriately also consider the features individually and combine them into additional advantageous combinations.

Shown are:

FIG. 1 a handheld power tool system comprising a handheld power tool in a side view,

FIG. 2 the handheld power tool system in a plan view,

FIG. 3 the handheld power tool system in a front view,

FIG. 4 the handheld power tool system in side view, comprising a handle housing shell,

FIG. 5 the handheld power tool system in a cross-sectional view in a separating plane of a handle housing of the handheld power tool,

FIG. 6 a top view of the handheld power tool in a cross-sectional view,

FIG. 7 the handheld power tool system in a placement and/or storage position on a work surface,

FIG. 8 the handheld power tool system in a further placement and/or storage position on the work surface,

FIG. 9A a perspective view of a base body of a protective device of the handheld power tool system,

FIG. 9B the base body of the protective device in a plan view,

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FIG. 10A a section of a drive housing of the handheld power tool in a perspective view,

FIG. 10B the drive housing in a plan view,

FIG. 11 a schematic sequence of a method for manufacturing a handheld power tool device for the handheld power tool, and

FIG. 12 a handheld power tool system with a handheld power tool in an alternative embodiment.

DETAILED DESCRIPTION

FIG. 1 shows a handheld power tool system 36a comprising a handheld power tool 50a, in particular a hand-held sander. The handheld power tool 50a is designed as a sander 10a, in particular as a battery-operated sander. Alternatively, however, it is also conceivable that the handheld power tool 50a, in particular the sander 10a, is designed as a grid-powered handheld power tool. The sander 10a is designed as a random orbital sander. Alternatively, however, it is also conceivable that the sander 10a is designed as an orbital sander, a delta sander, a flower sander, or the like.

The handheld power tool 50a comprises a tool holder 12a for holding a tool 178a. The tool holder 12a is designed as a sanding pad. The tool holder 12a is designed as a round sanding pad. Alternatively, however, it is also conceivable that the tool holder 12a is designed as a delta pad, as a square, preferably rectangular, sanding pad, or as another sanding pad that a skilled person considers useful.

The tool holder 12a comprises fastening means for fastening the tool 178a. The tool holder 12a preferably comprises a hook-and-loop surface 180a for fastening the tool 178a. Alternatively or additionally, it is conceivable that the fastening means comprise multiple hook-and-loop surfaces, one or multiple clips, a combination of these, or other fastening means that a skilled person considers useful. The tool 178a is designed as a sanding means, e.g. as an abrasive paper, an abrasive fleece, an abrasive grid, or the like.

The handheld power tool 50a comprises a drive unit 14a for driving the tool holder 12a. The drive unit 14a comprises a drive element 182a, preferably an eccentric (see FIG. 5). The drive element 182a is intended for driving about a drive axis 30a of the drive unit 14a. The handheld power tool 50a comprises an output unit 66a. The tool holder 12a is a portion of the output unit 66a. The output unit 66a comprises a connecting piece 90a. The connecting piece 90a is arranged, preferably fixed, on the drive element 182a. The tool holder 12a is fastened to the connecting piece 90a, in particular in a rotationally fixed manner, e.g. by means of a screw 184a or the like.

The output unit 66a comprises an output axis 186a. The output axis 186a preferably extends at least substantially parallel to the drive axis 30a. The drive axis 30a is at a distance from the output axis 186a. The connecting piece 90a and/or the tool holder 12a, in particular the output axis 186a, moves in at least one operating state about the drive axis 30a, preferably circularly. The drive element 182a is intended to drive the connecting piece 90a and/or the tool holder 12a, in particular the output axis 186a, to a, preferably circular, movement about the drive axis 30a of the drive element 182a.

The connecting piece 90a is rotatably attached to the drive element 182a, in particular rotatable about the output axis 186a. The handheld power tool 50a comprises at least one bearing element 188a. The connecting piece 90a is rotatably mounted on the drive element 182a via the bearing element 188a, in particular about the output axis 186a. The bearing element 188a is preferably arranged between the drive

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element 182a and the connecting piece 90a. The bearing element 188a can be designed as a radial bearing, in particular as a rolling bearing, e.g. as a ball bearing, or as a plain bearing.

The drive unit 14a is intended to drive the output unit 66a, in particular the tool holder 12a. The drive unit 14a comprises an electric motor 190a, in particular for driving the drive element 182a.

The handheld power tool 50a, in particular the sander 10a, comprises at least one further bearing element 300a. The drive element 182a is rotatably mounted on a drive housing 54a of the handheld power tool 50a via the further bearing element 300a, in particular rotatable about the drive axis 30a. The further bearing element 300a can be designed as a radial bearing, in particular as a rolling bearing, e.g. as a ball bearing, or as a plain bearing.

The handheld power tool 50, in particular the sander 10a, comprises a housing unit 16a. The housing unit 16a comprises a drive housing 54a for holding the drive unit 14a. The drive housing 54a is designed as a plastic housing. Alternatively, however, it is also conceivable that the drive housing 54a is at least partially designed as a metal housing, e.g. as an aluminum housing or the like.

The housing unit 16a comprises a handle housing 52a. The handle housing 52a is designed as a plastic housing. Alternatively, however, it is also conceivable that the handle housing 52a is at least partially designed as a metal housing, e.g. as an aluminum housing or the like. The handle housing 52a comprises two handle housing shells 192a, 194a, which are connected to one another in particular in a separating plane 146a of the handle housing 52a. The handle housing 52a is designed to be separate from the drive housing 54a, in particular arranged at a distance from the drive housing 54a. In particular, the handle housing 52a comprises a shell structure, with the two handle housing shells 192a, 194a acting as half shells.

The housing unit 16a comprises a bow-shaped handle 18a. The bow-shaped handle 18a is a portion of the handle housing 52a. The bow-shaped handle 18a comprises a finger grip surface 196a. The bow-shaped handle 18a comprises two walls 198a, in particular side walls, which delimit the finger grip surface 196a, in particular viewed in a main extension plane 42a of the bow-shaped handle 18a and in a direction perpendicular to the drive axis 30a. The bow-shaped handle 18a is designed as a closed bow-shaped handle. Alternatively, however, it is also conceivable that the bow-shaped handle 18a is designed as an open bow-shaped handle. The walls 198a define, in particular delimit, a handle recess 20a of the bow-shaped handle 18a at least partially.

A maximum longitudinal extent 22a of the handle recess 20a extends at least substantially parallel to a tool receiving surface 102a of the tool holder 12a. The maximum longitudinal extent 22a of the handle recess 20a extends in a main extension plane 64a of the handle housing 52a, preferably in the main extension plane 42a of the bow-shaped handle 18a, in particular in the separating plane 146a of the handle housing 52a. The tool holder surface 102a extends at least substantially perpendicular to the separating plane 146a, the main extension plane 64a of the handle housing 52a, and/or the main extension plane 42a of the bow-shaped handle 18a.

The main extension plane 42a of the bow-shaped handle 18a is in this case congruent with the main extension plane 64a of the handle housing 52a. Alternatively, it is conceivable that the main extension plane 42a of the bow-shaped handle 18a is different from the main extension plane 64a of the handle housing 52a. The separating plane 146a is in this case congruent with the main extension plane 64a of the

handle housing **52a**, in particular the main extension plane **42a** of the bow-shaped handle **18a**. Alternatively, however, it is also conceivable that the main extension plane **42a** of the bow-shaped handle **18a** and/or the main extension plane **64a** of the handle housing **52a** are/is different from the separating plane **146a**.

A ratio of the maximum longitudinal extent **22a** of the handle recess **20a** of the bow-shaped handle **18a** to a maximum longitudinal extent **24a** of the tool holder **12a** is at least 0.35, preferably at least 0.38. The ratio of the maximum longitudinal extent **22a** of the handle recess **20a** of the bow-shaped handle **18a** to a maximum longitudinal extent **24a** of the tool holder **12a** is at most 0.5, preferably at most 0.45, and preferably at most 0.42. Particularly preferably, the ratio of the maximum longitudinal extent **22a** of the handle recess **20a** of the bow-shaped handle **18a** to the maximum longitudinal extent **24a** of the tool holder **12a** is at least substantially 0.4.

The maximum longitudinal extent **22a** of the handle recess **20a** is at least 50 mm, preferably at least 55 mm, and particularly preferably at least 60 mm. The maximum longitudinal extent **22a** of the handle recess **20a** is at most 80 mm, preferably at most 70 mm, and particularly preferably at most 75 mm. The maximum longitudinal extent **22a** of the handle recess **20a** is in this case, by way of example, at least substantially 61 mm. Alternatively, however, it is also conceivable that the maximum longitudinal extent **22a** of the handle recess **20a** is greater than 80 mm or less than 50 mm.

The maximum longitudinal extent **24a** of the tool holder **12a** in this case corresponds in particular to a maximum diameter of the tool holder surface **102a**. The maximum longitudinal extent **24a** is in this case, by way of example, 150 mm, in particular between 148 mm and 152 mm. Alternatively, however, it is also conceivable that the maximum longitudinal extent **24a** of the tool holder **12a**, in particular the maximum diameter of the tool holder surface **102a**, is greater than 150 mm, in particular greater than 152 mm, or smaller than 150 mm, in particular smaller than 148 mm.

A maximum diameter of the tool **178a** is greater than the maximum longitudinal extent **24a** of the tool holder **12a**. Alternatively, however, it is also conceivable that the maximum diameter of the tool **178a** is the same size or smaller than the maximum longitudinal extent **24a** of the tool holder **12a**.

The maximum longitudinal extent **24a** of the tool holder **12a** extends at least substantially parallel to the maximum longitudinal extent **22a** of the handle recess **20a** and/or the tool holder surface **102a**. The maximum longitudinal extent **24a** of the tool holder **12a** extends along the separating plane **146a** of the handle housing **52a**, the main extension plane **64a** of the handle housing **52a**, and/or in the main extension plane **42a** of the bow-shaped handle **18a**. The maximum longitudinal extent **24a** of the tool holder **12a** is defined by a maximum longitudinal extent of the tool holder surface **102a**. The tool **178a** is intended to be arranged on the tool holder surface **102a**. The hook and loop surface **180a** corresponds to the tool holder surface **102a**.

The handle housing **52a** comprises a palm grip **160a**, in particular a grip knob. The palm grip **160a** is arranged such that the drive axis **30a** intersects the palm grip **160a**. In particular, the bow-shaped handle **18a** merges into the palm grip **160a**, preferably adjoining the palm grip **160a**.

The handheld power tool **50a** comprises an extraction fan **168a**. The extraction fan **168a** is intended for suction away from the workpiece during the machining of a workpiece and, in particular to blow it out of a dust outlet **44a**.

A fan housing **166a** of the housing unit **16a**, in particular of the drive housing **54a**, for holding the extraction fan **168a** is arranged between the tool holder **12a** and the palm grip **160a**. The handle housing **52a**, in particular the palm grip **160a**, surround a motor housing **78a** of the drive housing **54a**, preferably at least in a plane extending perpendicular to the drive axis **30a**.

The sander **10a** comprises a battery pack interface **218a** for electrical and/or mechanical connection to a battery pack **38a**. The battery pack interface **218a** includes guidance means, e.g. one or multiple guide rails or the like, for guiding the battery pack **38a** during an assembly and/or disassembly of the battery pack **38a** to and/or from the battery pack interface **218a**. The battery pack interface **218a**, in particular the guidance means, specifies an insertion direction for the battery pack **38a** when the battery pack **38a** is mounted on the battery pack interface **218a**. The direction of insertion preferably extends at least substantially perpendicular to the drive axis **30a**, to the separating plane **146a**, to the main extension plane **42a** of the bow-shaped handle **18a**, and/or to the main extension plane **64a** of the handle housing **52a**. The insertion direction extends at least substantially parallel to the tool holder surface **102a**.

The battery pack interface **218a** is arranged on the handle housing **52a**, in particular on a side of the bow-shaped handle **18a** facing away from the palm grip **160a**. The battery pack interface **218a** has no interface with the drive axis **30a** and/or with a main extension plane **232a** of the tool holder surface **102a**.

The handheld power tool **50a** comprises at least one input element **162a**. The handheld power tool **50a** can be switched on and/or off by an actuation of the input element **162a** by the operator. Alternatively or additionally, it is conceivable that a rotational speed of the handheld power tool **50a** or the like can be adjustable by way of the input element **162a**. The input element **162a** can, e.g. be designed as a button, a rotary wheel, a switch, or the like. The input element **162a** is arranged on the palm grip **160a**, preferably on a side of the palm grip **160a** facing away from the battery pack interface **218a**, the tool holder **12a** and/or the bow-shaped handle **18a**.

The sander **10a** is intended to be guided with one hand, in particular by means of the bow-shaped handle **18a** and the palm grip **160a**. Alternatively, however, it is also conceivable that the sander **10a** is designed as a two-hand guide, whereby one hand of the operator is arranged on the palm grip **160a** and one on the bow-shaped handle **18a**.

A center of gravity **26a** of the handheld power tool **50a** is arranged in the handle recess **20a**. The center of gravity **26a** is arranged within the handle recess **20a**, viewed at least in a direction perpendicular to the main extension plane **42a** of the bow-shaped handle **18a**, in particular to the main extension plane **64a** of the handle housing **52a**.

A minimum distance (not shown in this case) of the center of gravity **26a** from the separating plane **146a** of the handle housing **52a**, from the main extension plane **64a** of the handle housing **52a**, and/or from the main extension plane **42a** of the bow-shaped handle **18a** is smaller than a maximum transverse extent **200a** of the handle housing **52a**, starting from the separating plane **146a** of the handle housing **52a**, the main extension plane **64a** of the handle housing **52a**, and/or the main extension plane **42a** of the bow-shaped handle **18a**. It is also conceivable that the center of gravity **26a** is arranged in the separating plane **146a** of the handle housing **52a**, in the main extension plane **64a** of the handle housing **52a**, and/or in the main extension plane **42a** of the bow-shaped handle **18a**.

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The maximum transverse extent **200a** of the handle housing **52a**, in particular of the bow-shaped handle **18a**, in particular starting from the separating plane **146a** of the handle housing **52a**, the main extension plane **64a** of the handle housing **52a**, and/or the main extension plane **42a** of the bow-shaped handle **18a**, extends at least substantially perpendicular to the main extension plane **42a** of the bow-shaped handle **18a**, in particular to the main extension plane **64a** of the handle housing **52a**, to the separating plane **146a**, and/or to the drive axis **30a**. The maximum transverse extent **200a** of the handle housing **52a**, in particular of the bow-shaped handle **18a**, in particular starting from the separating plane **146a** of the handle housing **52a**, the main extension plane **64a** of the handle housing **52a**, and/or the main extension plane **42a** of the bow-shaped handle **18a**, extends at least substantially parallel to the tool holder surface **102a**.

A ratio of a maximum distance **28a** of the tool holder **12a** from the center of gravity **26a** in the direction of the drive axis **30a** of the drive unit **14a** to a minimum distance **32a** between the drive axis **30a** and the center of gravity **26a** is at least 2.8, preferably at least 3.0, preferably at least 3.1. The ratio of a maximum distance **28a** of the tool holder **12a** from the center of gravity **26a** in the direction of the drive axis **30a** of the drive unit **14a** to a minimum distance **32a** between the drive axis **30a** and the center of gravity **26a** is at most 3.5, preferably at most 3.2.

The maximum distance **28a** of the tool holder **12a** from the center of gravity **26a** in the direction of the drive axis **30a** is at least 30 mm, preferably at least 33 mm, preferably at least 35 mm. The maximum distance **28a** of the tool holder **12a** from the center of gravity **26a** in the direction of the drive axis **30a** is in particular at most 50 mm, preferably at most 45 mm and particularly preferably at most 40 mm. By way of example, the maximum distance **28a** of the tool holder **12a** from the center of gravity **26a** in the direction of the drive axis **30a** is at least substantially 38 mm here. Alternatively, however, it is also conceivable that the maximum distance **28a** of the tool holder **12a** from the center of gravity **26a** in the direction of the drive axis **30a** is less than 30 mm or greater than 50 mm.

The minimum distance **32a** of the drive axis **30a** from the center of gravity **26a** is in particular at least 8 mm, preferably at least 10 mm. The minimum distance **32a** of the drive axis **30a** from the center of gravity **26a** is in particular at most 20 mm, preferably at most 15 mm. By way of example, the minimum distance **32a** of the drive axis **30a** from the center of gravity **26a** is at least substantially 38 mm. Alternatively, however, it is also conceivable that the minimum distance **32a** of the drive axis **30a** from the center of gravity **26a** is less than 8 mm or greater than 20 mm.

The maximum distance **28a** of the tool holder **12a** from the center of gravity **26a** in the direction of the drive axis **30a** is defined by a distance of the tool holder surface **102a** from the center of gravity **26a** in the direction of the drive axis **30a**.

The drive axis **30a** extends at least substantially perpendicular to the tool holder surface **102a**. The drive axis **30a** extends at least substantially parallel to the main extension plane **64a** of the handle housing **52a**, in particular to the main extension plane **42a** of the bow-shaped handle **18a**, and/or to the separating plane **146a**. The drive axis **30a** extends along the main extension plane **64a** of the handle housing **52a**, along the main extension plane **42a** of the bow-shaped handle **18a**, and/or along the separating plane **146a**.

The minimum distance **32a** between the drive axis **30a** and the center of gravity **26a** is at least substantially per-

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pendicular to the drive axis **30a**. The minimum distance **32a** between the drive axis **30a** and the center of gravity **26a** extends at least substantially parallel to the tool holder surface **102a**, to the main extension plane **64a** of the handle housing **52a**, to the main extension plane **42a** of the bow-shaped handle **18a**, and/or to the separating plane **146a**.

A maximum height **34a** of the handheld power tool **50a**, in particular of the sander **10a**, has a ratio of at least 2.8, preferably at least 3.0, in the direction of the drive axis **30a** to the maximum distance **28a** between the tool holder **12a** and the center of gravity **26a** in the direction of the drive axis **30a**. The ratio of the maximum height **34a** of the sander **10a** in the direction of the drive axis **30a** to the maximum distance **28a** between the tool holder **12a** and the center of gravity **26a** in the direction of the drive axis **30a** is at most 3.2, preferably at most 3.1.

The maximum height **34a** of the sander **10a** is in this case, by way of example, at least 100 mm, preferably at least 110 mm. The maximum height **34a** of the sander **10a** is in this case, by way of example, at most 130 mm, preferably at most 120 mm. Alternatively, however, it is also conceivable that the maximum height **34a** of the sander **10a** in the direction of the drive axis **30a** is less than 110 mm or greater than 130 mm. The maximum height **34a** of the sander **10a** in the direction of the drive axis **30a** extends from the tool holder surface **102a** parallel to the drive axis **30a** to an outermost point of the handle housing **52a**.

A ratio of the maximum height **34a** of the sander **10a** to a maximum height **202a** of the handle recess **20a** in a direction extending parallel to the drive axis **30a** is at least 4.0, preferably at least 4.3, particularly preferably at least 4.4. The ratio of the maximum height **34a** of the sander **10a** to the maximum height **202a** of the handle recess **20a** extending in a direction parallel to the drive axis **30a** is at most 5.0, preferably at most 4.7, particularly preferably at most 4.6. The maximum height **34a** of the handle recess **20a** is in this case, by way of example, between 20 mm and 30 mm, preferably between 24 mm and 28 mm, and particularly preferably at least substantially 26 mm.

The handle recess **20a** is closed, preferably substantially ovally closed. The handle recess **20a** is closed when viewed in the main extension plane **42a** of the bow-shaped handle **18a**, in the main extension plane **64a** of the handle housing **52a**, and/or in the separating plane **146a**. The handle recess **20a** is closed by walls of the drive housing **54a**, in particular the fan housing **166a** of the drive housing **54a**, and the handle housing **52a**. Alternatively, it is conceivable that the handle recess **20a** is completely closed by walls of the handle housing **52a**.

The handle recess **20a** has an oval shape, in particular when viewed in a direction perpendicular to the main extension plane **42a** of the bow-shaped handle **18a**, preferably to the main extension plane **64a** of the handle housing **52a**. Alternatively, however, it is also conceivable that the handle recess **20a** has a rectangular shape, a square shape, a circular shape, or another shape that a skilled person considers useful, in particular when viewed in the direction perpendicular to the main extension plane **42a** of the bow-shaped handle **18a**, preferably to the main extension plane **64a** of the handle housing **52a**.

The handheld power tool system **36a** comprises the battery pack **38a** for supplying power to the handheld power tool **50a**. The battery pack **38a** has a battery pack center of gravity **40a**. In a state arranged on the sander **10a**, the battery pack center of gravity **40a** is offset from the main extension plane **42a** of the bow-shaped handle **18a**.

A minimum distance (not shown in this case) of the battery pack center of gravity **40a** from the separating plane **146a** of the handle housing **52a**, from the main extension plane **64a** of the handle housing **52a** and/or from the main extension plane **42a** of the bow-shaped handle **18a** is smaller than the maximum transverse extent **200a** of the handle housing **52a**, in particular of the bow-shaped handle **18a**, starting from the separating plane **146a** of the handle housing **52a**, the main extension plane **64a** of the handle housing **52a**, or the main extension plane **42a** of the bow-shaped handle **18a**.

The minimum distance of the battery pack center of gravity **40a** from the separating plane **146a** of the handle housing **52a**, from the main extension plane **64a** of the handle housing **52a**, and/or from the main extension plane **42a** of the bow-shaped handle **18a** is in particular less than 20 mm, preferably less than 15 mm, and preferably less than 10 mm. The minimum distance of the battery pack center of gravity **40a** from the separating plane **146a** of the handle housing **52a**, from the main extension plane **64a** of the handle housing **52a**, and/or from the main extension plane **42a** of the bow-shaped handle **18a** is in this case, by way of example, at least substantially 8 mm.

The handheld power tool system **36a** comprises a dust outlet **44a**. The dust outlet **44a** is arranged on the drive housing **54a**, in particular on the fan housing **166a**. The battery pack center of gravity **40a** and the dust outlet **44a** are arranged on different sides of the main extension plane **42a** of the bow-shaped handle **18a**. The center of gravity **26a** of the sander **10a** and the battery pack center of gravity **40a** are arranged on different sides of the main extension plane **42a** of the bow-shaped handle **18a**.

The handheld power tool system **36a** comprises a dust collection container **262a**, e.g. a dust box, a dust bag, or the like (see FIG. 2). The dust collection container **262a** can be attached, in particular detachably, to the dust outlet **44a**, e.g. by means of a screw connection, a latching connection, or the like. Alternatively, it is also conceivable that at least one dust outlet connecting piece of the dust collection container **262a** is designed to be integral with the dust outlet **44a**.

The dust collection container **262a** is rotatably mounted on the dust outlet **44a**.

Alternatively, however, it is also conceivable that the dust collection container **262a** can be connected to the dust outlet **44a** in a rotationally fixed manner. The dust collection container **262a** is intended for collecting dust and/or material removed by extraction, preferably by means of the suction fan **168a**.

Alternatively or additionally, it is conceivable that the dust outlet **44a** can be coupled to an active extraction device (not shown in this case), e.g. a vacuum cleaner or the like, in particular in the case of a handheld power tool **50a** which is designed without an extraction fan **168a**. The dust outlet **44a** is arranged on one side of the separating plane **146a**, the main extension plane **42a** of the bow-shaped handle **18a**, and/or the main extension plane **64a** of the handle housing **52a**.

A ratio of a minimum distance **46a** between the battery pack center of gravity **40a** in a state arranged on the sander **10a** and the drive axis **30a** to the maximum longitudinal extent **24a** of the tool holder **12a** is at least 0.85 and at most 1.2.

The minimum distance **46a** between the battery pack center of gravity **40a** in a state arranged on the sander **10a** and the drive axis **30a** extends at least substantially parallel to the tool holder surface **102a**, the main extension plane **42a** of the bow-shaped handle **18a**, the main extension plane **64a**

of the handle housing **52a**, and/or the separating plane **146a**. The minimum distance **46a** between the battery pack center of gravity **40a** in a state arranged on the sander **10a** and the drive axis **30a** extends at least substantially perpendicular to the drive axis **30a**.

By way of example, the minimum distance **46a** between the battery pack center of gravity **40a** and the drive axis **30a** is at least substantially 136 mm for a battery pack **38a** with 4 Ah electrical charge capacity, at least substantially 148 mm for a battery pack **38a** with 8 Ah electrical charge capacity and at least substantially 160 mm for a battery pack **38a** with 12 Ah electrical charge capacity.

The ratio of the minimum distance **46a** between the battery pack center of gravity **40a** in a state arranged on the sander **10a** and the drive axis **30a** to the maximum longitudinal extent **24a** of the tool holder **24a** is at least 0.85 and at most 0.95, preferably at least substantially 0.9, for a battery pack **38a** with 4 Ah electrical charge capacity.

The ratio of the minimum distance **46a** between the battery pack center of gravity **40a** in a state arranged on the sander **10a** and the drive axis **30a** to the maximum longitudinal extent **24a** of the tool holder **24a** is at least 0.95 and at most 1.05, preferably at least substantially 1.0, for a battery pack **38a** with 8 Ah electrical charge capacity.

The ratio of the minimum distance **46a** between the battery pack center of gravity **40a** in a state arranged on the sander **10a** and the drive axis **30a** to the maximum longitudinal extent **24a** of the tool holder **24a** is at least 1.0 and at most 1.1, preferably at least substantially 1.07, for a battery pack **38a** with 12 Ah electrical charge capacity.

An overall center of gravity **176a** of the handheld power tool system **36a** is arranged in the handle recess **20a** of the bow-shaped handle **18a**. The overall center of gravity **176a** of the handheld power tool system **36a** pertains to the handheld power tool **50a** with the battery pack **38a** and has no dust collection container **262a** and/or guard **120a**. A minimum distance (not shown in this case) of the overall center of gravity **176a** from the separating plane **146a** of the handle housing **52a**, from the main extension plane **64a** of the handle housing **52a** and/or from the main extension plane **42a** of the bow-shaped handle **18a** is smaller than the maximum transverse extent **200a** of the handle housing **52a**, in particular of the bow-shaped handle **18a**, starting from the separating plane **146a** of the handle housing **52a**, the main extension plane **64a** of the handle housing **52a**, and/or the main extension plane **42a** of the bow-shaped handle **18a**. It is conceivable that the overall center of gravity **176a** is arranged in the separating plane **146a** of the handle housing **52a**, in the main extension plane **64a** of the handle housing **52a**, and/or in the main extension plane **42a** of the bow-shaped handle **18a**.

A minimum distance **302a** of the drive axis **30a** from the overall center of gravity **176a** has a value of at least 55 mm to at least 70 mm, in particular depending on a weight of the battery pack **38a**. The minimum distance **302a** of the drive axis **30a** from the overall center of gravity **176a** has a value of at most 65 mm to at most 75 mm, in particular depending on the weight of the battery pack **38a**.

A maximum distance **304a** of the tool holder **12a** from the overall center of gravity **176a** in the direction of the drive axis **30a** has a value of at least 45 mm to at least 50 mm, in particular depending on the weight of the battery pack **38a**. The maximum distance **304a** of the tool holder **12a** from the overall center of gravity **176a** in the direction of the drive axis **30a** has a value of at most 52 mm to at most 55 mm, in particular depending on a weight of the battery pack **38a**.

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Preferably, the overall center of gravity **176a** of the handheld power tool system **36a** is arranged with at least one battery pack **38a** with a single-layer battery cell arrangement within the handle recess **20a** of the bow-shaped handle **18a**. A position of the overall center of gravity **176a** of the handheld power tool system **36a** with a battery pack **38a** with a single-layer battery cell arrangement is such that the handheld power tool **50a** can be placed on the tool holder **12a**, in particular without tilting.

By way of example, the minimum distance **302a** of the drive axis **30a** from the overall center of gravity **176a** is at least substantially 60 mm for a battery pack **38a** with 4 Ah of electrical charging capacity and at least substantially 72 mm for a battery pack **38a** with 8 Ah of electrical charging capacity.

By way of example, the maximum distance **304a** from the tool holder **12a** to the overall center of gravity **176a** in the direction of the drive axis **30a** is at least substantially 50 mm for a battery pack **38a** with 4 Ah electrical charging capacity and at least substantially 53 mm for a battery pack **38a** with 8 Ah electrical charging capacity.

A ratio of the minimum distance **46a** between the battery pack center of gravity **40a** in a state arranged on the sander **10a** and the drive axis **30a** to the maximum longitudinal extent **22a** of the handle recess **20a** of the bow-shaped handle **18a** is at least 2.0 and at most 3.0.

The ratio of the minimum distance **46a** between the battery pack center of gravity **40a** in a state arranged on the sander **10a** and the drive axis **30a** to the maximum longitudinal extent **22a** of the handle recess **20a** of the bow-shaped handle **18a** is at least 2.0 and at most 2.5, preferably at least substantially 2.2, for a battery pack **38a** with 4 Ah electrical charging capacity.

The ratio of the minimum distance **46a** between the battery pack center of gravity **40a** in a state arranged on the sander **10a** and the drive axis **30a** to the maximum longitudinal extent **22a** of the handle recess **20a** of the bow-shaped handle **18a** is at least 2.2 and at most 2.6, preferably at least substantially 2.4, for a battery pack **38a** with 8 Ah electrical charging capacity.

The ratio of the minimum distance **46a** between the battery pack center of gravity **40a** in a state arranged on the sander **10a** and the drive axis **30a** to the maximum longitudinal extent **22a** of the handle recess **20a** of the bow-shaped handle **18a** is at least 2.4 and at most 2.8, preferably at least substantially 2.6, for a battery pack **38a** with 12 Ah electrical charging capacity.

The handheld power tool **50a** comprises a handheld power tool device **48a**, in particular a handheld sander device. The handheld power tool device **48a** comprises the housing unit **16a**.

The drive housing **54a** defines the drive axis **30a** of the drive unit **14a**. In particular, the motor housing **78a** defines the drive axis **30a**. The motor housing **78a** comprises a motor holder, in the central longitudinal axis of which a motor axis **260a** comes to rest. The drive axis **30a** is defined by the motor axis **260a**. In particular, the central longitudinal axis is congruent with the drive axis **30a**. The drive housing **54a** is coupled to the handle housing **52a**.

The drive housing **54a** comprises a center bar **60a** on a surface **56a** of an upper side **58a** of the drive housing **54a**, which is raised relative to the surface **56a**. An end **62a** of the handle housing **52a** is arranged above the center bar **60a**. The center bar **60a** drops onto the surface **56a** in both directions, starting from a main extension plane **64a** of the handle housing **52a** and extending perpendicularly to the main extension plane **64a** of the handle housing **52a**.

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Two screw domes **280a** of the drive housing **54a** are arranged on side walls **278a** of the center bar **60a**. In particular, the center bar **60a** protrudes in a radial direction with respect to the drive axis **30a**, whereby a further screw dome **282a** of the drive housing **54a** is arranged in the protruding region of the center bar **60a**.

A maximum transverse extent **306a** of the center bar **60a** is smaller than a maximum total transverse extent **204a** of the handle housing **52a**. The maximum transverse extent **306a** of the center bar **60a** is at least substantially perpendicular to the main extension plane **42a** of the bow-shaped handle **18a**, to the main extension plane **64a** of the handle housing **52a**, to the separating plane **146a**, and/or to the drive axis **30a**. The maximum transverse extent **306a** of the center bar **60a** extends at least substantially parallel to the tool holder surface **102a** of the tool holder **12a**. The maximum transverse extent **306a** of the center bar **60a** is smaller than a maximum transverse extent **206a** of the handle housing **52a** at the end **62a** of the center bar **60a** arranged above the center bar **60a** (see FIG. 3). The end **62a** of the handle housing **52a** is arranged above the center bar **60a** with respect to the tool holder **12a** and/or the surface **56a**.

The center bar **60a** has a main extension axis **208a**. The main extension axis **208a** of the center bar **60a** extends at least substantially perpendicular to the drive axis **30a**. The main extension axis **208a** of the center bar **60a** extends at least substantially parallel to the main extension plane **42a** of the bow-shaped handle **18a**, to the main extension plane **64a** of the handle housing **52a**, to the separating plane **146a**, and/or to the tool holder surface **102a**. The main extension axis **208a** of the center bar **60a** extends in the main extension plane **42a** of the bow-shaped handle **18a**, in the main extension plane **64a** of the handle housing **52a** and/or in the separating plane **146a**.

The center bar **60a** extends, in particular at least in the direction of the main extension axis **208a** of the center bar **60a**, over a large portion of the surface **56a**, preferably over at least 50%, preferably over at least 75%, and particularly preferably over 90% of the surface **56a**. The upper side **58a**, preferably the surface **56a**, is arranged facing away from the tool holder **12a**. The center bar **60a** has a curvature in the longitudinal direction, in particular in the direction of the main extension axis **208a** of the center bar **60a**, preferably a curvature adapted to the surface **56a**. Alternatively, however, it is also conceivable that the center bar **60a** is designed with no curvature in the longitudinal direction.

The handle housing **52a** covers the center bar **60a** at least partially on both sides along an axis extending perpendicular to the main extension plane **64a** of the handle housing **52a**. The handle housing **52a** covers the center bar **60a** in the directions perpendicular to the main extension plane **64a** of the handle housing **52a** on both sides at a height of at least 10%. It is also conceivable that the handle housing **52a** largely covers the center bar **60a** on both sides at a height in the directions extending perpendicular to the main extension plane **64a** of the handle housing **52a**, in particular at least 50%, preferably at least 70%, and preferably at least 90%. The center bar **60a** is not covered by the handle housing **52a** in the longitudinal direction.

An end **62a** of the handle housing **52a** facing the center bar **60a**, in particular the end specified hereinabove, preferably an end **62a** substantially surrounding the drive housing **54a** in a ring-like manner, is arranged at a distance from the drive housing **54a**, in particular at a radial and axial distance. The end **62a** is arranged at a distance from the surface **56a**.

The motor housing **78a** is arranged on the center bar **60a**. The end **62a** surrounds the motor housing **78a** like a ring.

The motor housing **78a** is arranged on a side of the center bar **60a** facing away from the tool holder **12a**, in particular arranged above the center bar **60a** with respect to the tool holder **12a**. The motor housing **78a** is enclosed by the handle housing **52a** in a plane perpendicular to the drive axis **30a**. The motor housing **78a** is at least substantially completely enclosed by the handle housing **52a** in the radial direction starting from the drive axis **30a**. The motor housing **78a** is designed to be at least partially integral with the center bar **60a**. The handle housing **52a**, in particular the end **62a** of the handle housing **52a**, is arranged at a distance from the drive housing **54a**, in particular from the motor housing **78a**.

The end **62a** of the handle housing **52a** facing the center bar **60a** is adapted in shape to the drive housing **54a**, in particular the center bar **60a** and/or the surface **56a**. The end **62a** comprises two recesses **210a** for the center bar **60a**. It is conceivable that the end **62a** is at a constant distance from the drive housing **54a** in the direction of the drive axis **30a**. Alternatively, however, it is also conceivable that the distance between the end **62a** and the drive housing **54a** varies in the direction of the drive axis **30a**.

The center bar **60a** comprises a holder **68a** for a user interface **70a**. Alternatively or additionally, it is conceivable that the center bar **60a** comprises the holder **68a** for a communication module **72a**. The user interface **70a** and/or the communication module **72a** are/is a portion of the handheld power tool **50a**, in particular the sander **10a**, preferably the handheld power tool device **48a**.

The communication module **72a** can comprise one or multiple data transmission technologies, e.g. Bluetooth, NFC, or the like. Alternatively, it is conceivable that the handheld power tool **50a**, in particular the handheld power tool device **48a**, is designed without a communication module **72a**.

The holder **68a** is designed as a recess in the center bar **60a**. The holder **68a** is arranged on a side of the motor housing **78a** facing away from the bow-shaped handle **18a**. The holder **68a** is arranged on a side of the handle housing **52a** facing away from the battery pack interface **218a** of the handheld power tool **50a**. The battery pack interface **218a** is arranged on a side of the handheld power tool **50a**, in particular the sander **10a**, facing away from the user interface **70a**.

In this case, the user interface **70a** comprises, by way of example, two input elements **212a**. Alternatively, however, it is also conceivable that the user interface **70a** comprises only one input element **212a** or more than two input elements **212a**. The input elements **212a** of the user interface **70a** are in this case, by way of example, designed as buttons. Alternatively, it is conceivable that at least one of the input elements **212a** is designed as a rotary wheel, a switch, or the like. By way of example, the handheld power tool **50a** can be switched on and/or off via the user interface **70a**, in particular the input elements **212a**, a rotational speed of the handheld power tool **50a**, preferably of the sander **10a**, can be adjusted or the like.

The user interface **70a** in this case comprises, by way of example, five output elements **214a**. Alternatively, it is conceivable that the user interface **70a** comprises fewer or more than four output elements **214a**. The output elements **214a** are designed as lighting elements, in particular LEDs. Alternatively, it is conceivable that at least one of the output elements **214a** is designed as a display or the like. By way of example, the user interface **70a**, in particular via at least one of the output elements **214a**, can be used to output a rotational speed of the handheld power tool **50a**, preferably

of the sander **10a**, a charge status of the battery pack **38a**, a device status of the handheld power tool **50a**, or the like.

The user interface **70a** is connected by data technology to a control unit **216a** of the handheld power tool **50a**, in particular the sander **10a**, preferably in a wired and/or wireless manner. The control unit **216a** is intended to control the drive unit **14a**. In particular, the control unit **216a** comprises at least one processor and one memory element, as well as an operating program stored on the memory element. The memory element is designed as a digital storage medium, e.g. as a hard disk or the like. The control unit **216a** is arranged in the handle housing **52a**.

The communication module **72a** is configured to communicate with another device, e.g. a smartphone, a laptop, a server, a cloud, a gateway, or the like. The communication module **72a** is connected to the control unit **216a** and/or the user interface **70a** by data, in particular in a wireless and/or by wired manner.

The communication module **72a** is designed and in particular arranged separately from the user interface **70a**. The communication module **72a** is in this case, by way of example, arranged in the handle housing **52a**, in particular in the bow-shaped handle **18a**. The communication module **72a** is arranged between the motor housing **78a** and the battery pack interface **218a**. The communication module **72a** is arranged on a side of the handle housing **52a** on which the battery pack interface **218a** is arranged. It is also conceivable that the communication module **72a** is a portion of the control unit **216a**. Alternatively, it is conceivable that the communication module **72a** is arranged in a support **220a** of the handle housing **52a** on the drive housing **54a**, or in another position that a skilled person considers useful. Alternatively, it is also conceivable that the communication module **72a** is a portion of the user interface **70a**.

The handheld power tool **50a** comprises multiple cooling air inlet openings **266a**. Alternatively, it is conceivable that the handheld power tool **50a** only comprises a cooling air inlet opening **266a**. The cooling air inlet openings **266a** are in this case, by way of example, arranged on the handle housing **52a**, in particular on the bow-shaped handle **18a**. The cooling air inlet openings **266a** are arranged on a wall of the handle housing **52a** delimiting the handle recess **20a**. The cooling air inlet openings **266a** are arranged on a wall of the handle housing **52a** facing the tool holder **12a** and delimiting the handle recess **20a**. Alternatively, it is also conceivable that the cooling air inlet openings **266a** are arranged on a wall that is closest to the battery pack interface **218a** and delimits the handle recess **20a**. Alternatively, however, it is also conceivable that the cooling air inlet openings **266a** are arranged on the drive housing **54a** or at another position on the handle housing **52a** that a skilled person considers useful.

The handheld power tool **50a** comprises a motorized fan **268a**. The motor fan **268a** is arranged in the drive housing **54a**, in particular in the motor housing **78a**. The motor fan **268a** is intended for cooling the drive unit **14a**, in particular the electric motor **190a**. It is also conceivable that the control unit **216a** can be cooled by means of the motor fan **268a**. In particular, the engine fan **268a** is intended to draw in ambient air via the cooling air inlet openings **266a**. The motorized fan **268a** can be driven by the drive unit **14a**.

The drive housing **54a** comprises multiple air outlet openings **74a**, which are at least partially covered by the handle housing **52a**, in particular when viewed at least in a respective main outlet direction **76a** of the air outlet openings **74a**. Alternatively, however, it is conceivable that the drive housing **54a** only comprises one air outlet opening **74a**,

which is at least partially covered by the handle housing **52a**, in particular when viewed at least in a main outlet direction **76a** of the air outlet opening **74a**. The engine fan **268a** is intended to direct the air drawn in via the cooling air inlet openings **266a** via the drive unit **14a** to the air outlet openings **74a** and, in particular, to blow the engine exhaust air out of the air outlet openings **74a**.

The handle housing **52a** has no air outlet openings. The drive housing **54a**, in particular the motor housing **78a**, comprises cooling air inlet openings **270a**. The cooling air inlet ports **270a** of the engine housing **78a** are disposed on a side of the engine housing **78a** facing away from the tool holder **12a**. The cooling air inlet openings **270a** of the motor housing **78a** are arranged within the handle housing **52a**, in particular within the palm grip **160a**. The cooling air drawn in via the cooling air inlet openings **270a** of the motor housing **78a** can be introduced into the motor housing **78a** via the cooling air inlet openings **270a** on the handle housing **52a**.

The air outlet openings **74a** are at least partially covered by the handle housing **52a**, at least when viewed in a plane extending perpendicular to the drive axis **30a**. It is conceivable that the handle housing **52a** covers at least most of the air outlet openings **74a**, preferably at least 50%, preferably at least 75%, and particularly preferably at least 90%, in particular when viewed in a plane perpendicular to the drive axis **30a**. It is also conceivable that the handle housing **52a** completely covers the air outlet openings **74a**, in particular when viewed in a plane perpendicular to the drive axis **30a**. The air outlet openings **74a** are covered by the end **62a** of the handle housing **52a**, in particular when viewed in a plane extending perpendicular to the drive axis **30a**.

The motor housing **78a** comprises the air outlet openings **74a**. The air outlet openings **74a** are arranged above the center bar **60a** with respect to the tool holder **12a** and/or the surface **56a**. The air outlet openings **74a** are in this case arranged, by way of example, at a distance from one another in a circumferential direction, which in particular extends in a plane perpendicular to the drive axis **30a**.

The main outlet directions **76a** (in FIG. 4, only the main outlet direction **76a** of one of the air outlet openings **74a** is shown by way of example) are determined by an arrangement and/or geometry of the respective air outlet opening **74a**. The main outlet directions **76a** extend in this case, by way of example, at least substantially perpendicular to the drive axis **30a**. Alternatively, however, it is also conceivable that the main outlet directions **76a** extend at an angle to the drive axis **30a**, preferably inclined in the direction of the tool holder **12a**.

An opening **80a** for the escape of air from the air outlet openings **74a** is formed between the end **62a** of the handle housing **52a** facing the center bar **60a** and the drive housing **54a**. The opening **80a** is arranged opposite the upper side **58a**. The opening **80a** is formed by arranging the end **62a** of the handle housing **52a** at a distance from the driving housing **54a**, preferably the motor housing **78a**.

The drive housing **54a** comprises a closed cable channel **82a**. At least a part, in particular a large part, preferably at least 50%, preferably at least 70%, of the cable channel **82a** is covered by the handle housing **52a**, in particular when viewed in a plane extending perpendicular to the drive axis **30a**.

An open end **222a** of the cable channel **82a** is arranged at the holder **68a**. A further open end **224a** of the cable channel **82a** is arranged in an inner region **226a** of the handle housing **52a**. The cable channel **82a** extends from the holder **68a** into the inner region **226a** of the handle housing **52a**.

The cable channel **82a** extends, in particular in the direction of the drive axis **30a**, over the entire motor housing **78a**.

The handheld power tool **50a** and/or the handheld power tool device **48a** comprises an electrical connection unit (not shown in this case). The electrical connection unit comprises at least one electrical connection element, e.g. a cable. The electrical connection unit, in particular the electrical connection element, is intended to connect the user interface **70a** to the control unit **216a**, in particular at least in terms of data technology and/or control technology. The user interface **70a** can be supplied with electrical energy via the electrical connection unit, preferably via the at least one electrical connection element and/or a further electrical connection element of the electrical connection unit.

The cable channel **82a** is intended to hold at least a portion of the electrical connection unit, in particular at least the electrical connection element and/or the further electrical connection element. The electrical connection unit, in particular the at least one electrical connection element and/or the further electrical connection element, are/is arranged at least partially in the cable channel **82a**. The holder **68a** comprises at least one electrical connection (not shown in this case) for the user interface **70a**. At least the electrical connection element and/or the further electrical connection element is/are connected to the connection.

The drive housing **54a** comprises at least two housing shells **84a**, **86a**. The drive housing **54a** has a shell structure, with the two housing shells **84a**, **86a** acting as half shells. The two housing shells **84a**, **86a** of the drive housing **54a** are connected to one another in a drive housing separating plane **228a** of the drive housing **54a**. The drive housing separating plane **228a** extends at least substantially parallel to the separating plane **146a** of the handle housing **52a**. It is conceivable that the drive housing separating plane **228a** corresponds to the separating plane **146a** of the handle housing **52a**. Alternatively, however, it is also conceivable that the drive housing separating plane **228a** extends at an angle to separating plane **146a**.

The cable channel **82a** is arranged between the housing shells **84a**, **86a**. The cable channel **82a** extends within the outer walls of the housing shells **84a**, **86a**, in particular within at least one outer wall of one of the housing shells **84a**, **86a**. The cable channel **82a** adjoins a contact surface of the outer walls of the two housing shells **84a**, **86a**. A housing shell **84a** of the housing shells **84a**, **86a** comprises a groove **88a** for the cable channel **82a**. A further housing shell **86a** of the housing shells **84a**, **86a** forms a cover (not shown in this case) for the cable channel **82a**.

The cable channel **82a** is arranged at least partially on, in particular in, the center bar **60a**. It is conceivable that at least a portion of the cable channel **82a** extends through the center bar **60a**, e.g. at least 5%, 10%, or 20% of the cable channel **82a**.

The handheld power tool system **36a** comprises a support unit **92a**, which is designed for placing and/or storing the sander **10a** on a work surface **94a** in at least one placement and/or storage position, whereby the battery pack **38a** comprises a support surface **96a** of the support unit **92a** for the at least one placement and/or storage position. FIG. 7 shows the handheld power tool system **36a** in the placement and/or storage position.

The support surface **96a** is in this case, by way of example, a continuous surface, which in particular rests on the work surface **94a** in the at least one placement and/or storage position. The contact surface **96a** is in this case, by way of example, a flat surface. The work surface **94a** can, e.g., be a floor, a table, or the like. In particular, the battery

pack 38a comprises a battery pack housing 112a. The battery pack housing 112a comprises the contact surface 96a.

The battery pack 38a comprises a handheld power tool interface 230a for an electrical connection with the handheld power tool 50a, in particular the sander 10a. The handheld power tool interface 230a is intended to be arranged at the battery pack interface 218a. The support surface 96a is arranged on a side of the battery pack 38a facing away from the handheld power tool interface 230a.

In a state of the tool 178a arranged on the tool holder 12a, the tool holder 12a and/or the tool 178a are/is at a distance from the work surface 94a in the placement and/or storage position. In the placement and/or storage position, the sander 10a can only be stored via the support surface 96a. In the placement and/or storage position, the sander 10a is at a distance from the work surface 94a. In the placement and/or storage position, the sander 10a is connected to the work surface 94a via the battery pack 38a and can preferably be placed on the work surface 94a. In the placement and/or storage position, only the contact surface 96a on the battery pack 38a of the handheld power tool system 36a is in contact with the work surface 94a.

A main extension plane 98a of the support surface 96a extends to the main extension plane 232a of the tool holder surface 102a of the tool holder 12a at an angle 104a other than 90°. A main extension axis (not shown in this case) of the handheld power tool 50a, in particular of the sander 10a, extends at an angle other than 90° to the main extension plane 98a of the support surface 96a in the placement and/or storage position. The tool holder surface 102a extends at least substantially parallel to the main extension axis of the handheld power tool 50a, in particular of the sander 10a. Alternatively, it is conceivable that the tool holder surface 102a extends at an angle to the main extension axis of the handheld power tool 50a, in particular the sander 10a.

In particular, a handle axis 152a of the handle housing 52a extends at an angle other than 90° to the main extension plane 98a of the support surface 96a in the placement and/or storage position. The handle axis 152a is defined by the bow-shaped handle 18a. The handle axis 152a extends along the separating plane 146a and/or along the main extension plane 64a of the handle housing 52a, in particular along the main extension plane 42a of the bow-shaped handle 18a. The handle axis 152a corresponds to a main extension axis of a handle tube 234a of the bow-shaped handle 18a.

The main extension plane 232a of the tool holder surface 102a and the main extension plane 98a of the supporting surface 96a include an angle 104a which is greater than 80°. The angle 104a enclosed by the main extension plane 232a of the tool holder surface 102a and the main extension plane 98a of the supporting surface 96a has a value of less than 90° and greater than 80°. Particularly preferably, the angle 104a enclosed by the main extension plane 98a of the supporting surface 96a and the main extension plane 232a of the tool holder surface 102a has a value of at least substantially 86°.

The tool holder 12a, preferably the tool holder surface 102a, is inclined in the placement and/or storage position in a direction facing away from the work surface 94a. A normal 106a of the main extension plane 98a of the support surface 96a intersects the battery pack 38a and the center of gravity 26a of the sander 10a.

The support unit 92a is designed for placing and/or storing the sander 10a on the work surface 94a in at least one further placement and/or storage position, whereby the battery pack 38a comprises a further support surface 108a of

the support unit 92a for the further placement and/or storage position. FIG. 8 shows the handheld power tool system 36a in the further placement and/or storage position.

In the further placement and/or storage position, the tool holder surface 102a extends at least substantially parallel to a support plane 116a of the further placement and/or storage position and/or to the flat work surface 94a. In the further placement and/or storage position, the drive axis 30a extends at least substantially perpendicular to the support plane 116a of the further placement and/or storage position. An edge 110a of the battery pack housing 112a of the battery pack 38a forms the further support surface 108a. The edge 110a is arranged facing away from the tool holder 12a.

A main extension axis 118a of the battery pack 38a extends at least substantially perpendicular to the main extension axis of the sander 10a. The edge 110a of the battery pack housing 112a extends at least substantially perpendicular to the main extension axis of the handheld power tool 50a, in particular the sander 10a.

The sander 10a comprises only one support point 114a, which comes to rest in the support plane 116a of the further placement and/or storage position. The support point 114a is located on the handle housing 52a. Alternatively, it is also conceivable that the support point 114a is located on the drive housing 54a. The support point 114a of the sander 10a is arranged on a side of the handle housing 52a facing away from the tool holder 12a. The support point 114a is located on a surface 264a of the palm grip 160a. The palm grip 160a has a round shape. The surface 264a of the palm grip 160a, on which the support point 114a is arranged, is curved. Alternatively, however, it is also conceivable that the surface 264a of the palm grip 160a, on which the support point 114a is arranged, is flat.

In the further placement and/or storage position, the further support surface 108a and the support point 114a of the sander 10a are situated in the support plane 116a of the further placement and/or storage position. In the further placement and/or storage position, the handheld power tool system 36a is in contact with the work surface 94a via the further placement surface 108a on the battery pack 38a and via the support point 114a on the sander 10a.

The handheld power tool device 48a comprises the protective device 120a specified hereinabove, in particular a wall protection ring, for collision protection of the tool 178a, which can be arranged on the handheld power tool 50a, and/or the tool holder 12a with objects in a working environment of the handheld power tool 50a. The protective device 120a is designed in particular as a protective ring. The protective device 120a is preferably configured as a bumper for the handheld power tool 50a. In particular, the protective device 120a is intended to counteract a stop of the tool holder 12a and/or the tool 178a arranged on the tool holder 12a against walls or the like in the working environment.

The protective device 120a comprises a shutter unit 122a and/or a light guide unit 124a for at least one light source 132a.

The protective device 120a is arranged on the drive housing 54a in at least one operating state. The protective device 120a is intended to be arranged on the fan housing 166a. The shutter unit 122a is intended to shield a region from an incidence of light from the at least one light source 132a. The shutter unit 122a comprises multiple shutter elements 100a. Alternatively, it is conceivable that the shutter unit 122a comprises only one shutter element 100a. The shutter elements 100a are designed as opaque compo-

nents. It is also conceivable that the shutter elements **100a** are designed as reflectors, e.g. as mirrors or the like.

The protective device **120a** comprises a base body **236a**. The base body **236a** comprises multiple windows **276a**. The operator can look at the workpiece through the windows **276a** when working on the workpiece. Alternatively, however, it is also conceivable that the base body **236a** is designed without windows **276a**.

The shutter unit **122a**, in particular the shutter elements **100a**, are designed to be integral with the base body **236a**. The shutter unit **122a**, in particular the shutter elements **100a**, is/are formed by the base body **236a** of the protective device **120a**.

Alternatively, it is also conceivable that the shutter unit **122a**, in particular the shutter elements **100a**, is/are designed to be separate from the base body **236a**. In particular, it is alternatively conceivable that the shutter unit **122a**, preferably the shutter elements **100a**, is/are attachable to the base body **236a**, preferably detachably. It is also conceivable that the shutter unit **122a**, in particular the shutter elements **100a**, can be arranged adjustably on the base body **236a**, preferably in order to adjust a shielding region by the shutter unit **122a**.

The light guide unit **124a** is arranged on the base body **236a**. The light guide unit **124a** in this case comprises, by way of example, three light guide elements **238a**. Alternatively, it is conceivable that the light guide unit **124a** comprises more or less than three light guide elements **238a**. The light guide unit **124a** is intended to guide and/or emit light from the at least one light source **132a**. The light guide elements **238a** are, by way of example, designed as transparent fibers, tubes, or rods. The light guide elements **238a** are in this case, by way of example, designed as radiating elements, in particular as light-scattering elements, e.g. in the form of a roughening, a scattering glass, plastic or ceramic platelet, a scattering plastic film, or the like.

The shutter unit **122a** is intended to block a light emission of at least one light source **132a** at least in the direction of an intended operator side. The shutter unit **122a** is intended to block a light emission of the at least one light source **132a** in a direction facing away from the tool holder **12a**.

It is conceivable that the protective device **120a** comprises a light source unit **126a**. The light source unit **126a** comprises in this case, by way of example, two light sources **132a**. Alternatively, it is also conceivable that the light source unit **126a** comprises a different number of light sources **132a** from two, for example only one light source **132a** or more than two light sources **132a**. Furthermore, it is alternatively conceivable that the protective device **120a** is designed without a light source unit **126a**. The light source unit **126a**, in particular the light sources **132a**, is/are arranged on the base body **236a**. The light sources **132a** are, e.g., designed as LEDs, as incandescent lamps or as another light source that a skilled person considers useful. Alternatively, it is also conceivable that the handheld power tool **50a** comprises at least one of the light sources **132a**.

It is also conceivable that the protective device **120a** comprises at least one energy source **128a** for at least one light source **132a**. The protective device **120a** comprises in this case, by way of example, two energy sources **128a**. One of the energy sources **128a** is associated with each of the light sources **132a**. Alternatively, however, it is also conceivable that the protective device **120a** comprises only one energy source **128a** or more than two energy sources **128a** or is designed without energy sources **128a**. The energy sources **128a** are, e.g., designed as batteries, rechargeable batteries, or the like.

The energy sources **128a** are intended to supply energy to the respective light source **132a**, in particular the light source unit **126a**. The energy sources **128a** are connected to the light sources **132a**, in particular the light source unit **126a**, at least in one operating state. Alternatively or additionally, however, it is also conceivable that the light sources **132a**, in particular the light source unit **126a**, are/can be supplied with electrical energy via the handheld power tool **50a**, in particular the sander **10a**, for example via an electrical plug connection between the protective device **120a** and the handheld power tool **50a**.

The protective device **120a** in this case comprises, by way of example, an operating element **130a** for the light sources **132a**, in particular the light source unit **126a**. The at least one operating element **130a** is arranged on the base body **236a**. The at least one operating element **130a** can, e.g., be designed as a button, a rotary wheel, a switch, or the like. The at least one control element **130a** can, e.g., be used to switch the light sources **132a** on and/or off and/or to set a lighting parameter, e.g. a light color, an intensity, or the like.

Furthermore, it is conceivable that the protective device **120a** comprises a sensor unit **134a** for detecting an operation of the handheld power tool **50a**, in particular for activating at least one light source **132a** during operation of the handheld power tool **50a**.

The sensor unit **134a** is arranged on the base body **236a**. The sensor unit **134a** comprises at least one sensor element (not shown in this case), e.g. a vibration sensor or the like, for detecting operation of the handheld power tool **50a**, in particular the sander **10a**. The sensor unit **134a** comprises control electronics (not shown in this case) for processing the sensor signals. The control electronics are connected to the light sources **132a** for control purposes. The control electronics are intended to switch on the light sources **132a** when operation of the handheld power tool **50a** is detected. It is conceivable that the control electronics are a portion of the control unit **216a** of the hand-held power tool **50a**. Alternatively, however, it is also conceivable that the control electronics are designed to be separate from the control unit **216a**.

The protective device **120a** is at least partially made of a thermoplastic elastomer. The base body **236a** is at least partially made of a thermoplastic elastomer. At least one surface **240a** of the protective device **120a** facing away from the handheld power tool **50a**, in particular of the base body **236a**, is made of a thermoplastic elastomer.

It is also conceivable that the protective device **120a** comprises a protective layer consisting at least partially of a thermoplastic elastomer. The protective layer can be a cover placed over the base body **236a**, a layer glued to the base body **236a**, or the like. Alternatively or additionally, it is also conceivable that the protective layer is clamped, screwed, or the like onto the base body **236a**. Furthermore, it is also conceivable that the base body **236a** is overmolded by the protective layer, in particular that the protective layer is overmolded onto the base body **236a**, e.g. in a two-component injection molding process.

The protective device **120a** comprises a fastening unit **284a** for detachable fastening to the handheld power tool **50a** (see FIGS. 9A and 9B). The fastening unit **284a** comprises two pairs of fixing rails **286a** extending parallel to each other. The fixing rails **286a** each comprise a locking recess **288a**.

The handheld power tool **50a**, in particular the drive housing **54a**, preferably the fan housing **166a**, comprises a protective fastening unit **290a** corresponding to the fastening unit **284a** (see FIGS. 10A and 10B).

The protective mounting unit **290a** comprises two fixing rails **292a**. The fixing rails **292a** of the protective fastening unit **290a** are arranged between the fixing rails **286a** of the fastening unit **284a** in a state of the protective device **120a** arranged on the handheld power tool **50a**. The fixing rails **286a** of the fastening unit **284a** and the fixing rails **292a** of the protective fastening unit **290a** are intended for fixing in an axial direction relative to one another, preferably with respect to the drive axis **30a**. The protective fastening unit **290a** comprises two latching elements **294a**, in particular latching hooks or the like. The latching elements **294a** are intended to cooperate with the latching recesses **288a** of the fastening unit **284a** in order to fasten the protective device **120a** to the handheld power tool **50a**.

The protective device **120a** has an asymmetrical basic shape, in particular adapted to a dust outlet **44a** of the handheld power tool **50a**. A shape of the base body **236a** deviates from a circular shape along its longitudinal extent in at least one section, preferably in a region of the dust outlet **44a**.

The handheld power tool device **48a** comprises a damping unit **136a**, which couples the drive housing **54a** to the handle housing **52a**. The drive housing **54a** comprises an extension arm **138a** extending at least radially with respect to the drive axis **30a**. At least a portion **140a** of the damping unit **136a** is arranged on the extension arm **138a**. The housing shells **84a**, **86a** are screwed together on the extension arm **138a**, in particular via a screw dome **274a** of the housing shells **84a**, **86a**. A screw dome **272a** of the handle housing shells **192a**, **194a** in the region of the extension arm **138a** is arranged at a distance from the screw dome **274a** of the housing shells **84a**, **86a**.

The damping unit **136a** comprises multiple damping elements **164a**, **170a**, **246a**, **248a**. The damping elements **164a**, **170a**, **246a**, **248a** are made of at least one elastic material, in particular an elastic plastic. The damping elements **164a**, **170a**, **246a**, **248a** can be designed as blocks, rings, ring segments, plates, or the like.

At least a portion of the damping elements **164a**, **170a**, **246a**, **248a** are fixed between the drive housing **54a** and the handle housing **52a** by means of a clamp. It is also conceivable that the damping elements **164a**, **170a**, **246a**, **248a** comprise an adhesive layer for attachment to the drive housing **54a** and/or to the handle housing **52a**. Additionally or alternatively, it is conceivable that at least a portion of the damping elements **164a**, **170a**, **246a**, **248a** can be attached to the handle housing **52a** and/or the drive housing **54a** by means of damping element holders, e.g. recesses, holding bolts, or the like, and/or by means of a screw connection or the like.

The portion **140a** of the damping unit **136a** on the extension arm **138a** comprises, e.g., two damping elements **164a** of the damping elements **164a**, **170a**, **246a**, **248a**. Alternatively, it is also conceivable that the portion **140a** of the damping unit **136a** on the extension arm **138a** comprises only one damping element **164a** or more than two damping elements **164a**.

The drive housing **54a** is connected to the handle housing **52a** via the damping elements **164a**, **170a**, **246a**, **248a**. The damping unit **136a** is intended for vibration decoupling of the handle housing **52a** from the drive housing **54a**. The handle housing **52a** is coupled to the drive housing **54a** via two different coupling regions **250a**, **252a** of the damping unit **136a**. The handle housing **52a** comprises two inner regions **226a**, **254a** at a distance from one another, in each

of which a portion of the drive housing **54a** is arranged. The two coupling regions **250a**, **252a** are arranged in the two inner regions **226a**, **254a**.

The extension arm **138a** is arranged on a side of the drive housing **54a** facing the battery pack interface **218a**. The extension arm **138a** is arranged within the handle housing **52a**, in particular in a further inner region **254a** of the two inner regions **226a**, **254a** of the handle housing **52a**. The extension arm **138a** forms an outermost point of the drive housing **54a** when viewed in a direction perpendicular to the drive axis **30a** and extending along the separating plane **146a**. The extension arm **138a** is at least partially enclosed by the handle housing **52a**. It is also conceivable that the communication module **72a** is arranged between the extension arm **138a** and the battery pack interface **218a**, in particular in the further inner region **254a**.

The extension arm **138a** comprises at least one damping element holder **256a** for the damping elements **164a**. The shape of the damping elements **164a** is adapted to the damping element holder **256a**. The damping element holder **256a** comprises two holding bolts for holding one of the respective ring-shaped damping elements **164a**. The damping element holder **256a** is intended to hold the damping elements **164a** perpendicular to the main extension plane **64a** of the handle housing **52a**, in particular to the main extension plane **42a** of the bow-shaped handle **18a** and/or to the separating plane **146a**. The extension arm **138a** is, by way of example, in this case T-shaped, particularly in order to hold the at least two dampening elements **164a**.

Alternatively, however, it is also conceivable that, e.g., the damping element holder **256a** is designed as a recess, in particular for holding a damping element **164a** designed as a block, plate, or the like. The damping elements **164a** are fixed to the damping element holder **256a** via the handle housing **52a**, in particular the handle housing shells **192a**, **194a**. Alternatively or additionally, however, it is also conceivable that the damping elements **164a** are glued, screwed, or the like to the damping element holder **256a**.

A maximum distance **142a** of the extension arm **138a** from the drive axis **30a** is greater than a maximum distance **144a** of the tool holder **12a** from the drive axis **30a**, in particular when viewed along a separating plane **146a** of the handle housing **52a** (see FIG. 6). A maximum distance **258a** of the portion **140a** of the damping unit **136a** on the extension arm **138a** from the drive axis **30a** is greater than the maximum distance **144a** of the tool holder **12a** from the drive axis **30a**, in particular as viewed along the separating plane **146a** of the handle housing **52a**.

A ratio of a maximum transverse extent **148a** of the portion **140a** of the damping unit **136a** to a minimum distance **150a** of the portion **140a** of the damping unit **136a** from the handle axis **152a** of the handle housing **52a** is at least 0.9 and at most 1.05. The maximum transverse extent **148a** of the portion **140a** of the damping unit **136a** extends at least substantially perpendicular to the main extension plane **64a** of the handle housing **52a**, in particular the main extension plane **42a** of the bow-shaped handle **18a**, and/or the separating plane **146a**. The maximum transverse extent **148a** of the portion **140a** of the damping unit **136a** extends at least substantially parallel to the tool holder surface **102a**. The minimum distance **150a** of the portion **140a** of the damping unit **136a** from the handle axis **152a** is measured parallel to the separating plane **146a**, in particular to the main extension plane **42a** of the bow-shaped handle **18a**, preferably to the main extension plane **64a** of the handle housing **52a**.

The maximum transverse extent **148a** of the portion **140a** of the damping unit **136a** is in this case, by way of example, at least 40 mm, preferably at least 45 mm. The maximum transverse extent **148a** of the portion **140a** of the damping unit **136a** is in this case, by way of example, at most 60 mm, preferably at most 50 mm. The maximum transverse extent **148a** of the portion **140a** of the damping unit **136a** is defined by a maximum distance between the two damping elements **164a**, in particular in a direction perpendicular to the main extension plane **42a** of the bow-shaped handle **18a**.

A ratio of the maximum transverse extent **148a** of the portion **140a** of the damping unit **136a** to the maximum height **34a** of the sander **10a** is preferably at least 0.35, preferably at least 0.4. The ratio of the maximum transverse extent **148a** of the portion **140a** of the damping unit **136a** to the maximum height **34a** of the sander **10a** is preferably at most 0.5, preferably at most 0.45.

The minimum distance **150a** of the portion **140a** of the damping unit **136a** from the handle axis **152a** is preferably at least 45 mm, preferably at least 50 mm. The minimum distance **150a** of the portion **140a** of the damping unit **136a** from the handle axis **152a** is preferably at most 60 mm, preferably at most 55 mm. Alternatively, however, it is also conceivable that the minimum distance **150a** of the portion **140a** of the damping unit **136a** from the handle axis **152a** is less than 45 mm or greater than 60 mm.

A ratio of the maximum transverse extent **148a** of the portion **140a** of the damping unit **136a** to a minimum distance **154a** of the portion **140a** of the damping unit **136a** from the drive axis **30a** is at least 0.7, preferably at least 0.73. The ratio of the maximum transverse extent **148a** of the portion **140a** of the damping unit **136a** to the minimum distance **154a** of the portion **140a** of the damping unit **136a** from the drive axis **30a** is at most 0.8, preferably at most 0.75. The minimum distance **154a** of the portion **140a** of the damping unit **136a** from the drive axis **30a** is measured parallel to the separating plane **146a**, in particular to the main extension plane **42a** of the bow-shaped handle **18a**, preferably to the main extension plane **64a** of the handle housing **52a**.

The minimum distance **154a** of the portion **140a** of the damping unit **136a** from the drive axis **30a** is preferably at least 55 mm, preferably at least 60 mm. The minimum distance **154a** of the portion **140a** of the damping unit **136a** from the drive axis **30a** is preferably at most 70 mm, preferably at most 65 mm. Alternatively, however, it is also conceivable that the minimum distance **154a** of the portion **140a** of the damping unit **136a** from the drive axis **30a** is less than 55 mm or greater than 70 mm.

A ratio of a maximum height **156a** of the housing unit **16a** to the minimum distance **150a** of the portion **140a** of the damping unit **136a** on the extension arm **138a** from the handle axis **152a** of the handle housing **52a** is at most 2.25. The maximum height **156a** of the housing unit **16a** extends along the separating plane **146a**, in particular along the main extension plane **64a** of the handle housing **52a**, preferably along the main extension plane **42a** of the bow-shaped handle **18a**.

The maximum height **156a** of the housing unit **16a** extends parallel to the drive axis **30a**. The maximum height **156a** of the housing unit **16a** is in this case, by way of example, at least 90 mm, preferably at least 100 mm. The maximum height **156a** of the housing unit **16a** is in this case, by way of example, at most 120 mm, preferably at most 110 mm. Alternatively, however, it is also conceivable that the

maximum height **156a** of the housing unit **16a** in the direction of the drive axis **30a** is less than 90 mm or greater than 120 mm.

A ratio of a maximum extent **296a** of the damping unit **136a**, in particular a maximum distance **298a** of the damping elements **164a**, **170a**, **246a**, **248a**, in the direction of the drive axis **30a** to the maximum height **156a** of the housing unit **16a** is at least 0.8, preferably at least 0.85. A ratio of the maximum extent **296a** of the damping unit **136a**, in particular the maximum distance **298a** of the damping elements **164a**, **170a**, **246a**, **248a**, in the direction of the drive axis **30a** to the maximum height **34a** of the sander **10a** is at least 0.7, preferably at least 0.75. The maximum extent **296a** of the damping unit **136a**, in particular the maximum distance **298a** of the damping elements **164a**, **170a**, **246a**, **248a**, in the direction of the drive axis **30a** is in this case, by way of example, at least 85 mm, preferably at least 90 mm (see FIG. 4).

A ratio of the minimum distance **154a** of the portion **140a** of the damping unit **136a** from the drive axis **30a** in a direction perpendicular to the drive axis **30a** to a maximum longitudinal extent **158a** of the housing unit **16a** is greater than 0.3. The maximum longitudinal extent **158a** of the housing unit **16a** extends perpendicular to the drive axis **30a**. The maximum longitudinal extent **158a** of the housing unit **16a** extends at least substantially parallel to the tool holder surface **102a**. The maximum longitudinal extent **158a** of the housing unit **16a** extends along the separating plane **146a**, preferably along the main extension plane **64a** of the handle housing **52a**, in particular the main extension plane **42a** of the bow-shaped handle **18a**.

The maximum longitudinal extent **158a** of the housing unit **16a** is preferably at least 175 mm, preferably at least 180 mm. The maximum longitudinal extent **158a** of the housing unit **16a** is preferably at most 190 mm, preferably at most 185 mm. Alternatively, however, it is also conceivable that the maximum longitudinal extent **158a** of the housing unit **16a** is less than 175 mm or greater than 190 mm.

A ratio of the minimum distance **154a** between the portion **140a** of the damping unit **136a** and the drive axis **30a** to the minimum distance **150a** between the portion **140a** of the damping unit **136a** and the handle axis **152a** of the handle housing **52a** is at least 1.2 and at most 1.3.

The bow-shaped handle **18a** is supported via the portion **140a** of the damping unit **136a** and/or via the extension arm **138a** on the drive housing **54a** to form the closed handle recess **20a** of the bow-shaped handle **18a**.

The portion **140a** of the damping unit **136a** and/or the extension arm **138a** is arranged in the direction of the handle axis **152a** at the height of the handle recess **20a**. The portion **140a** of the damping unit **136a** on the extension arm **138a** and/or the extension arm **138a** is, viewed in the direction of the handle axis **152a**, arranged at least substantially completely within the maximum longitudinal extent **22a** of the handle recess **20a**.

The two damping elements **164a** of the portion **140a** of the damping unit **136a** on the extension arm **138a** are arranged at a distance from one another. The damping elements **164a** are arranged on opposite sides of the extension arm **138a**. The damping elements **164a** are arranged on mutually different sides of the separating plane **146a**, in particular the main extension plane **64a** of the handle housing **52a**, preferably the main extension plane **42a** of the bow-shaped handle **18a**. The damping elements **164a** are arranged mirror-symmetrically with respect to the separating plane **146a**, in particular the main extension plane **64a** of the

handle housing 52a, preferably the main extension plane 42a of the bow-shaped handle 18a.

The extension arm 138a is arranged on the fan housing 166a. The extension arm 138a is designed to be at least partially integral with the fan housing 166a. The extension arm 138a is arranged on a side of the fan housing 166a facing the battery pack interface 218a. The fan housing 166a is arranged between the motor housing 78a and the tool holder 12a. The extraction fan 168a is made of metal, in particular cast metal. Alternatively, however, it is also conceivable that the extraction fan 168a is made of plastic or a combination of plastic and metal.

The motor housing 78a and the fan housing 166a are at least partially designed to be integral with one another. The fan housing 166a and the motor housing 78a comprise common housing shells 84a, 86a. The two housing shells 84a, 86a each comprise one half of the fan housing 166a and one half of the motor housing 78a.

Two further damping elements 170a of the damping unit 136a are, e.g., arranged on the motor housing 78a such that the damping elements 170a are only partially enclosed by the handle housing 52a in a sectional plane 172a perpendicular to the drive axis 30a as viewed through the respective damping element 170a. Alternatively, however, it is also conceivable that the damping unit 136a comprises only one further damping element 170a or more than two further damping elements 170a.

The two further damping elements 170a are arranged between the handle housing 52a and the motor housing 78a. The two further damping elements 170a are arranged inside the handle housing 52a. The two further damping elements 170a are arranged on opposite sides of the motor housing 78a, in particular on different housing shells 84a, 86a. The two further damping elements 170a are arranged on different sides of the separating plane 146a and/or the drive housing separating plane 228a. The two further damping elements 170a are arranged symmetrically to the parting plane 146a and/or to the drive housing separating plane 228a. The two further damping elements 170a are designed to be cube-shaped. The further damping elements 170a are arranged at the height of the air outlet openings 74a when viewed in the axial direction.

The damping unit 136a comprises two sealing elements 174a, in particular a sealing ring in order to block an inlet of exhaust air into the handle housing 52a. The sealing elements 174a are arranged between the handle housing 52a and the drive housing 54a.

The sealing elements 174a are designed as circular arc segments. The sealing elements 174a are arranged inside the handle housing 52a. The sealing elements 174a are arranged on opposite sides of the motor housing 78a, in particular on different housing shells 84a, 86a. The sealing elements 174a are arranged on different sides of the separating plane 146a and/or the drive housing separating plane 228a. The sealing elements 174a are arranged symmetrically to separating plane 146a and/or to the drive housing separating plane 228a.

Alternatively, however, it is also conceivable that the damping unit 136a comprises only one sealing element 174a or more than two sealing elements 174a. The sealing elements 174a surround the motor housing 78a in a circumferential direction, which in particular extends in a plane perpendicular to the drive axis 30a, at least substantially completely, preferably in an angular range of at least 270°, preferably of at least 330°, and particularly preferably of at least 350°.

The sealing elements 174a are intended to at least substantially completely fill a gap between the motor housing 78a and the handle housing 52a. The sealing elements 174a are made of an elastic material, preferably an elastic plastic. The sealing elements 174a preferably have material hardness that is different from that of the damping elements 164a, 170a, 246a, 248a. The material hardness of the sealing elements 174a is lower than a material hardness of the damping elements 164a, 170a, 246a, 248a. Alternatively, it is conceivable that the sealing elements 174a and the damping elements 164a, 170a, 246a, 248a have an identical material hardness. The sealing elements 174a are arranged above the air outlet openings 74a with respect to the tool holder 12a. The sealing elements 174a are additionally intended to support and/or decouple the handle housing 52a from the drive housing 54a.

The damping unit 136a comprises four additional damping elements 246a. Alternatively, however, it is also conceivable that the damping unit 136a comprises fewer than four additional damping elements 246a or more than four additional damping elements 246a. The additional damping elements 246a are arranged between the handle housing 52a and the motor housing 78a. The additional damping elements 246a are arranged within the handle housing 52a. The additional damping elements 246a are arranged uniformly in a circumferential direction of the motor housing 78a, which in particular extends in a plane perpendicular to the drive axis 30a.

In each case, two of the additional damping elements 246a are arranged on opposite sides of the motor housing 78a, in particular on different housing shafts 84a, 86a. In each case, two of the additional damping elements 246a are arranged on different sides of the separating plane 146a and/or the drive housing separating plane 228a.

The additional damping elements 246a are arranged symmetrically to separating plane 146a and/or the drive housing separating plane 228a. The additional damping elements 246a are designed to be cube-shaped. The two sealing elements 174a are arranged between the further damping elements 170a and the additional damping elements 246a when viewed in the axial direction.

The damping unit 136a comprises a further additional damping element 248a. The further additional damping element 248a is intended for damping axial relative movements, in particular vibrations, between the drive housing 54a and the handle housing 52a with respect to the drive axis 30a and/or for axial support between the handle housing 52a and the drive housing 54a. Alternatively or additionally, however, it is also conceivable that the further additional damping element 248a is intended for damping radial movements and/or for radial support between the handle housing 52a and the drive housing 54a. The further additional damping element 248a is preferably arranged in an extension of the motor axis 260a, preferably between the motor housing 78a and the handle housing 52a.

The further additional damping element 248a is in this case, by way of example, designed in the shape of a circular cylinder. Alternatively, however, it is also conceivable that the further additional damping element 248a is designed to be cube-shaped or the like. The further additional damping element 248a is in this case, by way of example, arranged in a preloaded manner between the drive housing 54a and the handle housing 52a.

The maximum extent 296a of the damping unit 136a, in particular the maximum distance 298a of the damping elements 164a, 170a, 246a, 248a, in the direction of the drive axis 30a is defined by a maximum distance between

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the further additional damping element **248a** and the damping elements **164a** in the direction of the drive axis **30a**.

FIG. **11** shows a schematic sequence of a method for manufacturing a handheld power tool device **48a**, in particular the one specified hereinabove. The protective device **120a** is made at least partially of a thermoplastic elastomer.

In one method step, in particular a shaping step **242a**, the base body **236a** is shaped. In one method step, in particular an overmolding step **244a**, the base body **236a** is overmolded by a protective layer of thermoplastic elastomer. Alternatively, it is also conceivable that the base body **236a** is already at least partially molded by a thermoplastic elastomer, in particular during the molding step **242a**.

FIG. **12** shows a further exemplary embodiment of the invention. The following descriptions and the drawings are substantially limited to the differences between the exemplary embodiments, whereby reference can also be made in principle to the drawings and/or the description of the other exemplary embodiments, in particular FIGS. **1** to **11**, with regard to components with the same designation, in particular with regard to components having the same reference signs. To differentiate between the exemplary embodiments, the letter *a* is placed after the reference signs of the exemplary embodiment in FIGS. **1** to **11**. In the exemplary embodiment in FIG. **12**, the letter *a* is replaced by the letter *b*.

FIG. **12** shows a handheld power tool system **36b** comprising a handheld power tool **50b**. The handheld power tool **50b** is designed as a sander **10b**. The handheld power tool **50b** is designed as a battery-operated sander. Alternatively, however, it is also conceivable that the handheld power tool **50b**, in particular the sander **10b**, is designed as a grid-powered handheld power tool. The sander **10b** is designed as a random orbital sander. Alternatively, however, it is also conceivable that the sander **10b** is designed as an orbital sander, a delta sander, or the like.

The handheld power tool **50b** comprises a tool holder **12b**, in particular a sanding pad for holding a tool **178b**. The handheld power tool **50b** comprises a drive unit (not shown in this case) for driving the tool holder **12b**.

The handheld power tool **50b** comprises a housing unit **16b**.

The housing unit **16b** comprises a handle housing **52b** and a fan housing **166b** for holding an extraction fan. The fan housing **166b** is designed as a metal housing, in particular as an aluminum housing. Alternatively, however, it is also conceivable that the fan housing **166b** is at least partially designed as a plastic housing. The handle housing **52b** comprises a bow-shaped handle **18b**. The handle housing **52b** is coupled to the fan housing **166b**. A drive unit, in particular at least one electric motor, of the handheld power tool **50b** is arranged in the handle housing **52b**.

A ratio of a maximum longitudinal extent of a handle recess **20b** of the bow-shaped handle **18b** to a maximum longitudinal extent of the tool holder **12b** is at least 0.35 and at most 0.5. The bow-shaped handle **18b** is designed to be closed. Viewed in the main extension plane of the bow-shaped handle **18b**, the handle recess **20b** is enclosed, in particular delimited, by walls of the handle housing **52b**. The fan housing **166b** lies directly against the handle housing **52b**. The handle housing **52b** comprises air outlet openings **74b**.

The handheld power tool **50b** comprises a handheld power tool device **48b**. The housing unit **16b** is a portion of the handheld power tool device **48b**. The fan housing **166b** comprises a center bar **60b** on a surface of an upper side of the fan housing **166b** that is raised relative to the surface.

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The center bar **60b** drops onto the surface in both directions starting from a main extension plane (not shown in this case) of the handle housing **52b** and extending perpendicularly to the main extension plane of the handle housing **52b**.

The handheld power tool system **36b** comprises a battery pack **38b** for supplying power to the handheld power tool **50b**. The handheld power tool system **36b** comprises a support unit **92b**, which is designed for placing and/or storing the sander **10b** on a work surface (not shown in this case) in at least one placement and/or storage position, whereby the battery pack **38b** comprises a support surface of the support unit **92b** for the at least one placement and/or storage position.

The handheld power tool **50b** comprises a holder **68b** for a user interface **70b** of the handheld power tool **50b**. The holder **68b** is limited by the handle housing **52b** and the fan housing **166b**.

The invention claimed is:

1. A handheld power tool device for a sander, comprising: a housing unit including a handle housing and a drive housing configured to hold a drive unit, the drive housing defining a drive axis of the drive unit; and a damping unit configured to couple the drive housing to the handle housing, wherein the drive housing comprises an extension arm configured to extend, at least radially with respect to the drive axis and on which at least a portion of the damping unit is arranged.
2. The handheld power tool device according to claim 1, further comprising: a tool holder configured to hold a tool, wherein a maximum distance of the extension arm from the drive axis is greater than a maximum distance of the tool holder from the drive axis, along a separating plane of the handle housing.
3. The handheld power tool device according to claim 1, wherein a ratio between a maximum transverse extent of the portion of the damping unit and a minimum distance of the portion of the damping unit from a handle axis of the handle housing is at least 0.9 and at most 1.05.
4. The handheld power tool device according to claim 1, wherein a ratio between a maximum transverse extent of the portion of the damping unit and a minimum distance of the portion of the damping unit from the drive axis is at least 0.7 and at most 0.8.
5. The handheld power tool device according to claim 3, wherein a ratio between a maximum height of the housing unit and the minimum distance of the portion of the damping unit on the extension arm from the handle axis of the handle housing is at most 2.25.
6. The handheld power tool device according to claim 1, wherein a ratio between a minimum distance of the portion of the damping unit from the drive axis in a direction perpendicular to the drive axis and a maximum longitudinal extent of the housing unit is greater than 0.3.
7. The handheld power tool device according to claim 1, wherein a ratio between a minimum distance between the portion of the damping unit and the drive axis and a minimum distance between the portion of the damping unit and a handle axis of the handle housing is at least 1.2 and at most 1.3.
8. The handheld power tool device according to claim 1, wherein the handle housing comprises a bow-shaped handle supported by the portion of the damping unit and/or the extension arm on the drive housing in order to form a closed handle recess of the bow-shaped handle.

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9. The handheld power tool device according to claim 8, wherein the portion of the damping unit and/or the extension arm is arranged in a direction of the handle axis at a height of the handle recess.

10. The handheld power tool device according to claim 1, wherein the portion of the damping unit arranged on the extension arm comprises at least two damping elements arranged at a distance from one another.

11. The handheld power tool device according to claim 1, wherein:

the housing unit comprises a fan housing configured to hold extraction fan, and

the extension arm is arranged on the fan housing.

12. The handheld power tool device according to claim 1, wherein:

the drive housing comprises a motor housing configured to hold the drive unit and a fan housing configured to hold an extraction fan, and

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the motor housing and the fan housing are configured to be least partially integral with one another.

13. The handheld power tool device according to claim 1, wherein the drive housing comprises a motor housing, on which at least one damping element of the damping unit is arranged, such that the damping element is only partially enclosed by the handle housing through the damping element in a sectional plane perpendicular to the drive axis.

14. The handheld power tool device according to claim 1, wherein:

the damping unit comprises at least one sealing element configured to block an inlet of exhaust air into the housing unit, and

the at least one sealing element is arranged between the handle housing and the drive housing.

15. A sander comprising:

a handheld power tool device according to claim 1.

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